National Technical University of Athens

School of Electrical and Computer Engineering

Modeling Smart Electric Vehicle Charging

A Multi-Level Intelligence Approach with Renewable Energy Integration

Diploma Thesis

by

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Vehicle-to-Grid (V2G)...

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Abstract

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This diploma thesis investigates the application of smart electric vehicle (EV) charging systems in office buildings with renewable energy integration. Different levels of intelligence are analyzed, ranging from simple rule-based approaches to advanced optimization algorithms and Vehicle-to-Grid (V2G) technology...

Keywords: Electric vehicles, smart charging, renewable energy, optimization, Vehicle-to-Grid, office buildings

Acknowledgments

[Write your acknowledgments here]

I would like to express my sincere gratitude to my supervisors Alexandros Flamos and Dimitris Fotopoulos for their guidance and support throughout this research...

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Chapter 1

Introduction

1.1 Background and Motivation

The global transition toward sustainable transportation has accelerated the adoption of electric vehicles (EVs). As EV penetration increases, workplace charging infrastructure in office buildings becomes increasingly important. However, uncontrolled charging can create significant challenges:

- Peak demand stress on the electrical grid
- High electricity costs during peak hours
- Underutilization of on-site renewable energy generation
- Grid capacity constraints and potential infrastructure upgrades

Smart charging systems offer a solution by intelligently coordinating EV charging with building energy consumption, renewable energy production, and dynamic electricity pricing. Office buildings present an ideal use case due to predictable occupancy patterns and potential for solar photovoltaic (PV) integration.

1.2 Smart Charging Levels

This thesis investigates multiple levels of charging intelligence, each building upon the previous:

1.2.1 Level 0: Baseline (Uncontrolled Charging)

EVs charge immediately upon arrival at maximum available rate without any optimization or coordination. This represents the current common practice and serves as the baseline for comparison.

Characteristics:

- No intelligence or optimization
- Maximum charging rate whenever connected
- No consideration of electricity prices or renewable availability
- Simple implementation but potentially high costs

1.2.2 Level 1: Simple Rule-Based Charging

Basic heuristic algorithms that follow predefined rules for charging decisions.

Characteristics:

- Priority-based allocation (first-come, first-served)
- Preferential use of building solar surplus
- Respect for grid capacity constraints
- Low computational requirements
- Predictable behavior

1.2.3 Level 2: Intelligent Optimization

Advanced optimization algorithms that learn or compute optimal charging schedules:

- **Reinforcement Learning (RL):** Q-learning algorithm that learns optimal charging policies through trial and error interaction with the environment

1.2.4 Level 3: Vehicle-to-Grid (V2G)

Bidirectional energy flow enabling EVs to discharge energy back to the building/grid during high-demand or high-price periods.

Characteristics:

- Bidirectional charging infrastructure
- Revenue generation from energy arbitrage
- Grid services and demand response capability
- Additional constraints to preserve EV battery and ensure mobility needs

1.3 Role of Renewable Energy

Solar photovoltaic systems on office buildings can significantly reduce charging costs and grid dependence. Key considerations include:

- Temporal alignment: Solar production peaks during midday when many EVs are parked
- Cost reduction: On-site generation eliminates grid purchase costs
- Grid relief: Reduces peak demand on distribution infrastructure
- Sustainability: Lower carbon footprint for EV charging

The challenge lies in optimally coordinating EV charging with variable solar production while meeting mobility requirements and respecting grid constraints.

1.4 Benefits of Smart Charging in Office Buildings

Office buildings present unique advantages for smart EV charging:

1. **Predictable Patterns:** Regular arrival/departure times enable better optimization
2. **Long Parking Duration:** 8-10 hour parking windows provide flexibility for scheduling
3. **Solar Alignment:** Peak solar production coincides with parking hours
4. **Centralized Management:** Single entity controls both building and charging infrastructure
5. **Economic Incentives:** Reduced operational costs benefit building owners/tenants

1.5 Literature Review

1.5.1 Smart EV Charging

[Review key papers on smart EV charging]

Recent research has explored various approaches to EV charging optimization. [1] demonstrated that coordinated charging can reduce costs by 30-40% compared to uncontrolled charging...

1.5.2 Optimization Algorithms

[Review papers on RL for EV charging]

Reinforcement learning has shown promise in handling uncertain and dynamic environments [2]...

1.5.3 Vehicle-to-Grid Technology

[Review V2G literature]

V2G technology enables bidirectional energy flow, allowing EVs to provide grid services [3]. Studies have shown potential revenue of €200-500 per vehicle annually...

1.5.4 Renewable Integration

[Review papers on solar+EV integration]

Integration of solar PV with EV charging has been studied extensively [4]. Key findings indicate that smart charging can increase solar self-consumption by 40-60%...

1.6 Research Objectives

The primary objectives of this thesis are:

1. **Model Development:** Develop mathematical models for different intelligence levels (0-3) of EV charging systems in office buildings
2. **Algorithm Implementation:** Implement and compare optimization approaches: Simple rule-based and Reinforcement Learning (Q-Learning)
3. **Performance Evaluation:** Quantify the economic benefits, renewable energy utilization, and constraint satisfaction for each approach
4. **V2G Assessment:** Evaluate the additional value provided by Vehicle-to-Grid capability
5. **Practical Recommendations:** Provide guidance on which intelligence level offers the best cost-benefit ratio for office building applications

1.7 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 - Methodology:** Presents the mathematical models and algorithms for each intelligence level

- **Chapter 3 - Case Study:** Describes the office building scenario, data sources, and simulation setup
- **Chapter 4 - Results:** Presents simulation results with comparative analysis
- **Chapter 5 - Conclusions:** Summarizes findings and provides recommendations for future work

Chapter 2

Methodology

This chapter describes the model and algorithm for the different parts of the System that is responsible for the EV charging. It includes components, optimizations and implementation for the different optimization levels.

2.1 System Components

2.1.1 Office Building Model

The office building is modeled by it's energy consumption and it's renewable energy generation.

Energy Consumption

The building's hourly energy consumption profile $C_b(t)$ is derived from the ELMAS dataset (European Load Model for Aggregated Sectors), specifically the Office cluster (#5) from `Time_series_18_clusters.csv`.

Let L_t be the hourly energy load of the Office cluster base series ($T = 8{,}736$ hours for year 2018) with yearly energy:

$$E_{cluster} = \sum_{t=1}^{T} L_t \tag{2.1}$$

To represent a single office building, the per-customer annual energy is calculated for each class c belonging to the Office cluster column:

$$e_c = \frac{E_c}{N_c} \tag{2.2}$$

where E_c is the annual energy and N_c the number of customers in class c. The required annual energy is set to the 75th percentile of the customer distribution:

$$E_{target} = Q_{0.75}(\{e_c\}) = 58{,}831 \text{ kWh} \tag{2.3}$$

The energy profile is then scaled to match this target:

$$L_t^{scaled} = L_t \cdot \frac{E_{target}}{E_{cluster}} \tag{2.4}$$

To introduce noise to the hourly energy profile, two multiplicative noise factors are applied—a daily factor α_d (constant within each day) and an hourly factor β_t:

$$\alpha_d \sim \mathcal{N}(1, \sigma_d^2), \quad \sigma_d = 0.08, \quad \alpha_d \in [0.7, 1.3] \tag{2.5}$$

$$\beta_t \sim \mathcal{N}(1, \sigma_h^2), \quad \sigma_h = 0.03, \quad \beta_t \in [0.9, 1.1] \tag{2.6}$$

The synthetic series before re-normalization:

$$L_t^{syn} = L_t^{scaled} \cdot \alpha_d(t) \cdot \beta_t \tag{2.7}$$

Finally, the series is re-scaled to exactly preserve the target annual energy:

$$C_b(t) = L_t^{syn} \cdot \frac{E_{target}}{\sum_{t=1}^{T} L_t^{syn}} \tag{2.8}$$

The single-office profile has a mean load of 6.73 kW and a peak load of 11.92 kW.

Solar Photovoltaic System

The solar PV system creates electricity based on irradiance following the equation:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) \tag{2.9}$$

where:

- A_{panel} = solar panel area (m^2)
- η_{panel} = panel efficiency
- $I(t)$ = solar irradiance at hour t (W/m^2)

Building Battery Storage

The building has a battery with capacity C and SOC $\text{SoC}_b(t)$.

The max discharge power at time t is bounded by both the efficiency and the available energy above the minimum SoC:

$$P_{dis,t} = \min(P_{max},\ (\text{SoC}_{t-1} - \text{SoC}_{min}) \cdot \eta) \tag{2.10}$$

Similarly, the max charge is bounded by the efficiency and the remaining capacity:

$$P_{ch,t} = \min(P_{max},\ C - \mathrm{SoC}_{t-1}) \tag{2.11}$$

The state of charge depends on whether the building has extra or missing energy. When extra solar energy is available, the battery charges:

$$\mathrm{SoC}_t = \mathrm{SoC}_{t-1} + \min(P_{ch,t},\ E_{excess}(t)) \tag{2.12}$$

When the building demand overcomes the supply, the battery discharges:

$$\mathrm{SoC}_t = \mathrm{SoC}_{t-1} - \min\left(\frac{P_{dis,t}}{\eta},\ \frac{E_{missing}(t)}{\eta}\right) \tag{2.13}$$

where:

- P_{max} = maximum charge/discharge power [kW]
- C = battery capacity [kWh]
- SoC_{min} = minimum allowable state of charge
- η = round-trip efficiency
- $E_{excess}(t) = \max(0,\ P_{solar}(t) - C_b(t))$ = surplus solar energy
- $E_{missing}(t) = \max(0,\ C_b(t) - P_{solar}(t))$ = energy deficit

Constraints:

$$\mathrm{SoC}_{min} \leq \mathrm{SoC}_t \leq C \tag{2.14}$$

$$0 \leq P_{ch,t} \leq P_{max} \tag{2.15}$$

$$0 \leq P_{dis,t} \leq P_{max} \tag{2.16}$$

Net Energy Demand

The building's net energy demand (positive = needs grid, negative = surplus):

$$E_{net}^{building}(t) = C_b(t) - P_{solar}(t) - E_{discharge}^{b}(t) + E_{charge}^{b}(t) \tag{2.17}$$

2.1.2 Electric Vehicle Fleet Model

Individual EV Characteristics

Each EV i in the fleet has the following info:

- Battery capacity: B_{cap}^{i} [kWh]

- State of charge: $\mathrm{SoC}^i(t)$ [dimensionless, 0-1]
- Maximum charge rate: $P^i_{max,charge}$ [kW]
- Maximum discharge rate: $P^i_{max,discharge}$ [kW] (for V2G)
- Arrival time: $t^i_{arrival}$ [hour]
- Departure time: t^i_{target} [hour]
- Desired SoC: $\mathrm{SoC}^i_{desired}$ [dimensionless]

EV Battery Dynamics

The state of charge evolves for EV i following the equation:

$$\mathrm{SoC}^i(t+1) = \mathrm{SoC}^i(t) + \frac{\eta_{charge} \cdot E^i_{charge}(t) - E^i_{discharge}(t)/\eta_{discharge}}{B^i_{cap}} \tag{2.18}$$

where η_{charge} and $\eta_{discharge}$ are charging and discharging efficiencies (typically 0.95).

Constraints

Each EV must satisfy:

$$\mathrm{SoC}_{min} \leq \mathrm{SoC}^i(t) \leq 1.0 \quad \forall t \tag{2.19}$$

$$\mathrm{SoC}^i(t^i_{target}) \geq \mathrm{SoC}^i_{desired} \tag{2.20}$$

$$E^i_{charge}(t) = 0 \quad \text{if } t < t^i_{arrival} \text{ or } t \geq t^i_{target} \tag{2.21}$$

Constraint (2.20) makes sure that each EV reaches its desired charge level before departure.

2.1.3 Grid Model

Dynamic Pricing

The grid uses daily pricing following the equation:

$$\pi_{buy}(t) = \text{price\_profile}[t \bmod 24] \quad [€/\text{kWh}] \tag{2.22}$$

For V2G scenarios, the sell price:

$$\pi_{sell}(t) = \text{sell\_price\_profile}[t \bmod 24] \quad [€/\text{kWh}] \tag{2.23}$$

Typically, $\pi_{sell}(t) \leq \pi_{buy}(t)$ (80-90% of purchase price).

Grid Capacity Constraint

The total grid power consumption has a constraint:

$$E_{grid}^{building}(t) + \sum_{i=1}^{N_{EV}} E_{grid}^{i}(t) \leq P_{grid,max}(t) \tag{2.24}$$

where $P_{grid,max}(t)$ represents the hourly grid capacity limit.

2.2 Optimization Problem Formulation

2.2.1 Objective Function

The goal is to minimize the total daily electricity cost:

$$\min \sum_{t=0}^{T-1} \left[\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot E_{V2G}^{total}(t)\right] \tag{2.25}$$

where:

- $E_{grid}^{total}(t)$ = total energy purchased from grid at hour t
- $E_{V2G}^{total}(t)$ = total energy sold back to grid (V2G only)
- T = simulation duration (24 hours)

2.2.2 Energy Balance

For each hour t, the energy equation must be satisfied:

$$C_b(t) + \sum_{i=1}^{N_{EV}} E_{charge}^{i}(t) = P_{solar}(t) + E_{grid}^{total}(t) + \sum_{i=1}^{N_{EV}} E_{discharge}^{i}(t) \tag{2.26}$$

2.3 Level 0: Baseline (Uncontrolled Charging)

2.3.1 Algorithm Description

In the baseline scenario, each EV charges at max rate upon arrival:

$$E_{charge}^{i}(t) = \begin{cases} P_{max,charge}^{i} & \text{if } t_{arrival}^{i} \leq t < t_{target}^{i} \text{ and } \mathrm{SoC}^{i}(t) < 1.0 \\ 0 & \text{otherwise} \end{cases} \tag{2.27}$$

2.3.2 Mathematical Formulation

The total cost for charging:

$$Cost_{Level0} = \sum_{t=0}^{T-1} \pi_{buy}(t) \cdot \left[\max(0, E_{net}^{building}(t)) + \sum_{i=1}^{N_{EV}} E_{charge}^{i}(t) \right] \tag{2.28}$$

Key Assumptions:

- No coordination between EVs
- Solar energy used for building first, remainder for EVs if available

2.4 Level 1: Simple Rule-Based Charging

2.4.1 Algorithm Description

The simple algorithm follows a priority-based heuristic:

Algorithm 1 Simple Rule-Based Charging

for each hour t **do**
  Calculate building surplus: $E_{surplus}(t) = \max(0, P_{solar}(t) - C_b(t))$
  Calculate available grid capacity: $P_{available}(t) = P_{grid,max}(t) - \max(0, C_b(t) - P_{solar}(t))$
  $E_{remaining} \leftarrow E_{surplus}(t)$
  for each EV i (in order) **do**
    if $t_{arrival}^{i} \leq t < t_{target}^{i}$ and $\mathrm{SoC}^{i}(t) < \mathrm{SoC}_{desired}^{i}$ **then**
      $E_{needed}^{i} = \min(P_{max,charge}^{i}, (\mathrm{SoC}_{desired}^{i} - \mathrm{SoC}^{i}(t)) \cdot B_{cap}^{i})$
      $E_{from\_surplus} = \min(E_{needed}^{i}, E_{remaining})$
      $E_{from\_grid} = \min(E_{needed}^{i} - E_{from\_surplus}, P_{available}(t))$
      $E_{total} = E_{from\_surplus} + E_{from\_grid}$
      Charge EV i with E_{total}
      Update $E_{remaining}$ and $P_{available}(t)$
    end if
  end for
end for

2.4.2 Mathematical Formulation

Energy for EV i at hour t:

$$E_{charge}^{i}(t) = \min\left(P_{max,charge}^{i},\ (\mathrm{SoC}_{desired}^{i} - \mathrm{SoC}^{i}(t)) \cdot B_{cap}^{i},\ E_{available}(t)\right) \tag{2.29}$$

where $E_{available}(t)$ contains both building extra energy and remaining grid capacity.

2.5 Level 2: Intelligent Optimization Algorithms

2.5.1 Reinforcement Learning (Q-Learning)

State Space

The state representation for EV i at hour t:

$$s^i(t) = (\text{EV\_id},\ \text{SoC}^i_{bin}(t),\ t,\ \text{grid\_availability}) \tag{2.30}$$

where $\text{SoC}^i_{bin}(t)$ is the discretized SoC (bins of 0.1).

Action Space

For charge-only systems:

$$\mathcal{A} = \{\text{charge}, \text{standby}\} \tag{2.31}$$

For V2G-enabled systems:

$$\mathcal{A}_{V2G} = \{\text{charge}, \text{discharge}, \text{standby}\} \tag{2.32}$$

Reward Function

The reward function is the following:

$$\begin{aligned}
R(s, a, s') = & -\alpha_1 \cdot E^i_{grid}(t) \cdot \pi_{buy}(t) \quad \text{(minimize grid cost)} \\
& + \alpha_2 \cdot E^i_{solar}(t) \quad \text{(maximize solar usage)} \\
& + \alpha_3 \cdot \mathbb{1}[\text{SoC}^i(t^i_{target}) \geq \text{SoC}^i_{desired}] \quad \text{(target achievement)} \\
& - \alpha_4 \cdot \max(0, \text{SoC}^i_{desired} - \text{SoC}^i(t^i_{target})) \quad \text{(penalty for shortfall)} \\
& - \alpha_5 \cdot \mathbb{1}[\text{SoC}^i(t) < \text{SoC}_{min}] \quad \text{(constraint violation)}
\end{aligned} \tag{2.33}$$

where $\alpha_1, \ldots, \alpha_5$ are weights.

Q-Learning Update Rule

The Q-value is updated using the temporal difference method:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[R(s, a, s') + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \tag{2.34}$$

where:

- α = learning rate (0.1-0.15)
- γ = discount factor (0.95-1.0)

- s' = next state after taking action a in state s

Policy Execution

After training, the policy selects actions greedily:

$$a^*(s) = \arg\max_{a \in \mathcal{A}} Q(s, a) \tag{2.35}$$

2.6 Level 3: Vehicle-to-Grid (V2G)

2.6.1 Bidirectional Energy Flow

V2G expands the optimization to include discharge decisions:

$$E^i_{net}(t) = E^i_{charge}(t) - E^i_{discharge}(t) \tag{2.36}$$

2.6.2 V2G Constraints

Additional constraints for V2G are the following:

$$E^i_{discharge}(t) \leq (\mathrm{SoC}^i(t) - \mathrm{SoC}^i_{desired}) \cdot B^i_{cap} \quad \text{(preserve target SoC)} \tag{2.37}$$

$$E^i_{charge}(t) \cdot E^i_{discharge}(t) = 0 \quad \text{(no simultaneous charge/discharge)} \tag{2.38}$$

2.6.3 Revenue Calculation

Total benefit with V2G:

$$\mathrm{Benefit}_{V2G} = -\sum_{t=0}^{T-1} \left[\pi_{buy}(t) \cdot E^{total}_{grid}(t) - \pi_{sell}(t) \cdot \sum_{i=1}^{N_{EV}} E^i_{discharge}(t) \right] \tag{2.39}$$

Positive benefit indicates revenue or cost saving.

2.6.4 V2G Integration with Algorithms

The Q-Learning algorithm is extended to handle V2G by:

1. Expanding action space to include discharge
2. Modifying reward/cost functions to include sell reward
3. Adding discharge constraints to preserve battery state

2.7 Key Assumptions and Limitations

2.7.1 Assumptions

1. **Perfect Forecasting:** Solar production and building consumption are known in advance
2. **Deterministic Arrivals:** EV arrival/departure times are fixed (within random windows)
3. **No Battery Degradation:** Charging/discharging cycles do not degrade battery capacity
4. **Fixed Efficiency:** Battery charging efficiency is constant at 95%
5. **Ideal Grid Connection:** No transmission losses or power quality issues
6. **Price-Taker:** EV charging does not affect electricity prices

2.7.2 Limitations

1. Simplified battery model (linear SoC dynamics)
2. No modeling of battery thermal effects
3. Single-day simulation
4. No consideration of user preferences or comfort
5. Computational complexity and time not analyzed in detail

Chapter 3

Case Study: Office Building Scenario

This chapter describes the office building scenario, that is used to evaluate the smart charging algorithms. Data souces are provided along with parameter selection and justification for the chosen values.

The case study simulates a typical office building with:

- Rooftop solar PV installation
- EV charging infrastructure for 15 vehicles
- Grid connection with capacity constraints
- 24-hour simulation period

3.1 Building Energy Profile

3.1.1 Energy Consumption Data

Data Source

The building energy consumption profile is derived from the ELMAS (European Load Model for Aggregated Sectors) dataset, specifically the Office cluster (#5):

- Dataset: `office_single_building_timeseries.csv`
- Source: ELMAS dataset – `Time_series_18_clusters.csv` (Office cluster #5)
- Building type: Office/commercial
- Target annual energy: 58,831 kWh (75th percentile of per-customer distribution)
- Temporal resolution: Hourly (8,736 data points for year 2018)

The profile was created by scaling the cluster load shape to a single-building target energy and adding stochastic variability (daily $\sigma_d = 0.08$, hourly $\sigma_h = 0.03$) to introduce realistic fluctuations, followed by re-normalization to preserve the annual energy total (see Section 2.1.1 for full methodology).

Profile Characteristics

Figure 3.1 shows the hourly consumption pattern:

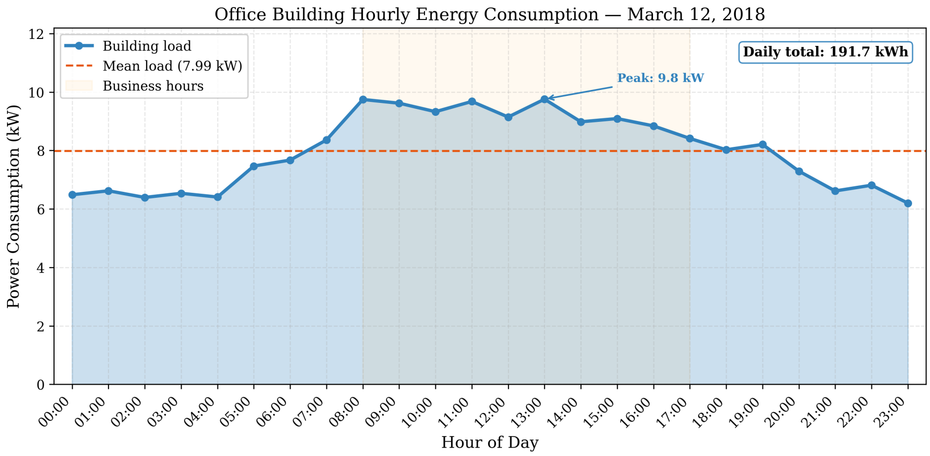

Figure 3.1: Office building hourly energy consumption profile

Annual statistics:

- Mean load: 6.73 kW
- Peak load: 11.92 kW (business hours)
- Annual energy: 58,831 kWh
- Daily consumption: ∼161 kWh (annual / 365)

Validation

The consumption profile comes from the ELMAS source data:

- The per-customer annual energy distribution (mean 10,429 kWh, P75 = 58,831 kWh) refers to small to medium-sized office buildings
- The 75th percentile was chosen to represent a medium-to-large office
- The cluster load gives realistic hourly patterns (working-hours peak, overnight baseline)
- Stochastic noise is used such that the profile is not artificially smooth while preserving the correct total energy

3.1.2 Solar PV System

System Specifications

The rooftop solar PV system parameters:

Table 3.1: Solar PV system specifications

Parameter	Value
Panel area	150 m^2
Panel efficiency	20%
Peak power	30 kW
Technology	Monocrystalline PERC (JA Solar DeepBlue 3.0 Pro)
Reference module	JAM54S31-405/MR (405 Wp, 20.7% efficiency)
Number of modules	74

The reference module is the JA Solar DeepBlue 3.0 Pro (JAM54S31-405/MR) [5], a widely deployed monocrystalline PERC half-cut cell panel rated at 405 Wp with a module efficiency of 20.7% (STC). The installation comprises 74 units, yielding a combined unit area of approximately 144.5 m^2 ($\approx$ 150 m^2) and a peak system power of 29.97 kW ($\approx$ 30 kW).

Irradiance Data

Solar irradiance data is obtained from PVGIS (Photovoltaic Geographical Information System):

- Source: European Commission Joint Research Centre PVGIS database [6]
- Location: Paris, France (48.857°N, 2.352°E, elevation 32 m)
- Radiation database: PVGIS-SARAH3
- Panel orientation: South-facing (azimuth 0°), tilt 37° (site-optimum)
- Date: March 12, 2018 (representative spring day)
- Temporal resolution: Hourly
- Data file: `irradiance_pvgis.csv`

Solar Production Profile

The hourly solar power production:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) = 150\,\mathrm{m}^2 \cdot 0.20 \cdot I(t) \tag{3.1}$$

Figure 3.2 illustrates the production profile:

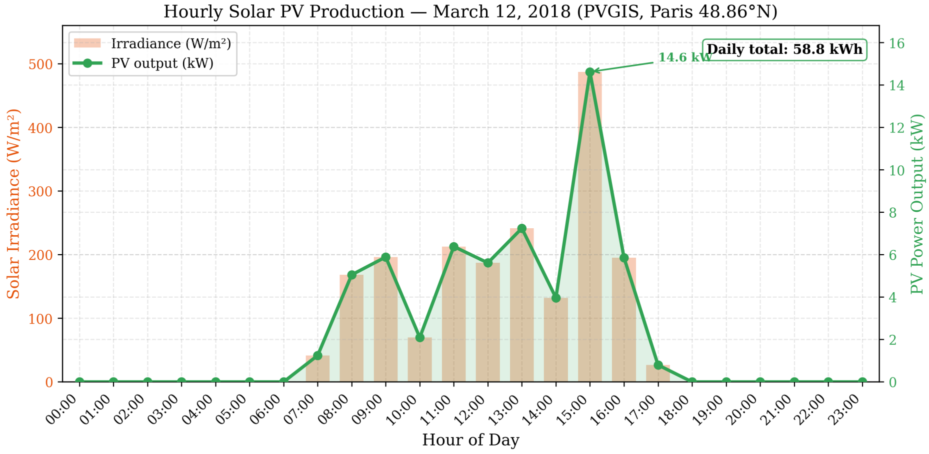

Figure 3.2: Hourly solar PV production profile (March 12, 2018)

Key observations:

- Peak production: ∼14.6 kW at 15:00 (peak irradiance of 487 W/m^2)
- Total daily production: 58.8 kWh
- Solar production partially offsets building consumption during business hours (7:00–17:00), with a brief surplus at 15:00
- Temporal alignment with EV parking hours (6 AM – 6 PM)
- Variable irradiance pattern due to sporadic cloud cover typical of Paris in spring

Selection Justification

March 12, 2018 was selected because:

- Spring day with moderate solar production (∼59 kWh)
- Realistic variable-cloud conditions typical of the Paris region, providing a realistic test of solar integration without unrealistic clear-sky irradiance
- Typical seasonal pattern (not extreme summer/winter)
- Validated satellite-derived irradiance data available from PVGIS (SARAH3 database)

3.1.3 Building Battery Storage

The building includes a stationary battery storage system:

Table 3.2: Building battery specifications

Parameter	Value
Capacity	80 kWh
Efficiency	95%
Initial SoC	50%
Depth of discharge limit	80%
Max charge rate	50 kW
Max discharge rate	50 kW
Chemistry	Lithium Iron Phosphate (LFP)
Reference product	BYD Battery-Box Premium HVM

The battery parameters are representative of a modular Lithium Iron Phosphate (LFP) system based on the BYD Battery-Box Premium HVM series [7]. The HVM system uses cobalt-free LFP cells with a certified round-trip efficiency of $\geq 96\%$ (HTW Berlin) and offers modular scalability in 2.76 kWh increments. In the simulated configuration, the equivalent of four HVM 19.3 towers ($4 \times 19.32 = 77.3$ kWh ≈ 80 kWh) provides a combined maximum charge/discharge power of approximately 50 kW.

3.2 Electric Vehicle Fleet

3.2.1 Fleet Composition

The simulation supports a fleet of 15 electric vehicles, representing a typical office building with 50-100 employees (assuming 15-30% EV adoption).

Vehicle Specifications

Each EV is modeled with identical technical characteristics:

Table 3.3: Electric vehicle specifications

Parameter	Value
Battery capacity	100 kWh
Reference vehicle	Tesla Model 3 Long Range
Max charge rate	22 kW (Level 2 AC charger)
Max discharge rate	22 kW (V2G capable)
Energy consumption	0.18 kWh/km
Depth of discharge limit	90%
Minimum SoC	20%

Justification

- **100 kWh capacity:** Representative of modern long-range EVs (Tesla Model 3, Hyundai Ioniq 5, etc.)
- **22 kW charging:** Standard Level 2 AC charging (common in workplace installations)
- **0.18 kWh/km:** Typical energy consumption for mid-size EVs
- **20% minimum SoC:** Prevents deep discharge and battery degradation

3.2.2 Mobility Patterns

Arrival and Departure Times

The EV fleet follows typical office worker patterns:

Table 3.4: EV mobility parameters

Parameter	**Value**
Arrival window	6:00 - 9:00 AM
Departure window	4:00 - 8:00 PM
Parking duration	7-14 hours
Initial SoC range	20-30%
Desired SoC range	70-80%

Each EV's specific arrival and departure times are randomly sampled from the specified windows to simulate realistic variability.

State of Charge Requirements

- **Initial SoC (20-30%):** Represents battery level after morning commute (assuming 30-50 km commute)
- **Desired SoC (70-80%):** Ensures sufficient charge for evening commute plus reserve (100-150 km range)

Justification

The mobility pattern reflects typical office workers:

- Morning arrival spread (6-9 AM) accounts for flexible work hours
- Evening departure spread (4-8 PM) reflects varied work schedules
- Long parking duration provides flexibility for smart charging

- SoC requirements ensure mobility needs are met

3.3 Grid Parameters

3.3.1 Electricity Pricing

Day-Ahead Market Prices

The simulation uses real wholesale day-ahead electricity prices from the EPEX SPOT exchange for the French bidding zone on March 12, 2018. The data was gathered from the ENTSO-E Transparency Platform [8] and the Fraunhofer ISE Energy-Charts API [9].

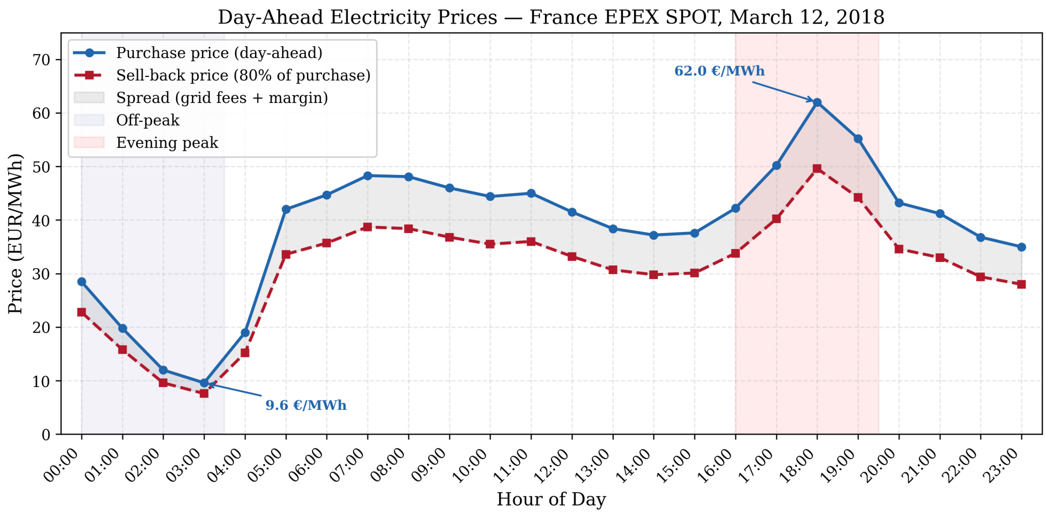

Figure 3.3: Hourly electricity purchase and sell-back prices (France EPEX SPOT, March 12, 2018)

Table 3.5 summarizes the hourly day-ahead prices by time period. The complete hourly data is provided in Appendix B.2.

Table 3.5: Day-ahead electricity prices by time period (France, March 12, 2018)

Time Period	Purchase (EUR/MWh)	Purchase (€/kWh)	Sell-back (€/
Night (00:00–03:00)	9.56–28.46	0.010–0.028	0.008–0.02
Morning ramp (04:00–07:00)	19.04–48.34	0.019–0.048	0.015–0.03
Business hours (08:00–11:00)	44.41–48.05	0.044–0.048	0.036–0.03
Midday (12:00–15:00)	37.24–41.50	0.037–0.042	0.030–0.03
Evening peak (16:00–19:00)	42.23–62.00	0.042–0.062	0.034–0.05
Night (20:00–23:00)	35.04–43.23	0.035–0.043	0.028–0.03
Daily average	**38.67**	**0.039**	**0.031**

Price Profile Characteristics

The day-ahead price profile for March 12, 2018 exhibits characteristic features of the French electricity market:

- **Off-peak trough (02:00–04:00):** Prices drop to 9.56–12.00 EUR/MWh due to low demand and high nuclear baseload generation
- **Morning ramp (05:00–08:00):** Prices rise sharply from 19.04 to 48.34 EUR/MWh as commercial and industrial loads come online
- **Daytime plateau (08:00–15:00):** Prices stabilize at 37–48 EUR/MWh; solar PV and hydro generation partially offset demand
- **Evening peak (18:00):** The daily maximum of 62.00 EUR/MWh occurs at hour 19 (6–7 PM) as solar production decreases while residential heating and lighting demand peaks
- **Evening decline (20:00–23:00):** Prices gradually fall from 43.23 to 35.04 EUR/MWh as demand subsides; France's large nuclear fleet provides ample overnight capacity

Sell-Back Prices (V2G)

For the V2G scenario, sell-back prices are set at 80% of the day-ahead purchase price. This spread accounts for:

- Grid access and balancing fees
- Aggregator margin
- Round-trip efficiency losses

Data Source

The electricity price data was obtained from the ENTSO-E Transparency Platform [8] and cross-validated against the Fraunhofer ISE Energy-Charts API [9]. The original market data is from EPEX SPOT SE, the power exchange operating the French day-ahead auction as part of the Single Day-Ahead Coupling (SDAC) mechanism [10].

3.3.2 Grid Capacity Constraints

Capacity Limits

The grid connection has hourly capacity limits:

$$P_{grid,max}(t) = 80\,\text{kW} \quad \forall t \tag{3.2}$$

Justification

The 80 kW limit represents:

- Typical distribution transformer capacity for small commercial buildings
- Building baseline consumption: ∼8 kW
- Available for EV charging: ∼70 kW (sufficient for 3-4 simultaneous Level 2 chargers)
- Creates realistic constraint requiring intelligent coordination

Without smart charging, 15 EVs at 22 kW each would require 330 kW, far exceeding the 80 kW limit. This constraint necessitates intelligent scheduling.

3.4 Simulation Configuration

3.4.1 Time Parameters

Table 3.6: Simulation time parameters

Parameter	Value
Simulation duration	24 hours
Time step	1 hour
Simulation date	March 12, 2018
Start time	00:00 (midnight)
End time	23:00 (11 PM)

3.4.2 Algorithm Parameters

The optimization algorithm requires specific hyperparameters:

Reinforcement Learning

Table 3.7: Q-Learning hyperparameters

Parameter	Charge-Only	V2G
Training episodes	8,000	15,000
Learning rate (α)	0.15	0.15
Discount factor (γ)	0.99	1.0
Exploration rate (ϵ)	0.35	0.10
SoC discretization	0.1 (10% bins)	0.1 (10% bins)

Justification:

- Higher episodes for V2G due to increased complexity (3 actions vs 2)
- $\gamma = 1.0$ for V2G: Future rewards equally important (long-term planning)
- Lower ϵ for V2G: More exploitation after learning optimal policy

3.5 Data Sources Summary

Table 3.8: Summary of data sources

Data Type	Source	Reference
Building consumption	ELMAS dataset (Office cluster #5)	[11]
Solar irradiance	PVGIS	[6]
Electricity prices	EPEX SPOT / ENTSO-E (France)	[8], [10]
Solar PV modules	JA Solar DeepBlue 3.0 Pro	[5]
Building battery	BYD Battery-Box Premium HVM	[7]
EV specifications	Tesla Model 3	[12]
Charging standards	IEC 61851	[13]

3.6 Simulation Scenarios

3.6.1 Scenario 1: Charge-Only (No V2G)

Configuration: `multi_ev_config.yml`

This scenario evaluates unidirectional charging with:

- 15 EVs with charge-only capability
- Algorithms: Simple, RL (Q-Learning)
- Grid capacity: 80 kW
- No discharge allowed

Purpose: Establish baseline performance for intelligent charging without V2G.

3.6.2 Scenario 2: V2G-Enabled

Configuration: `multi_ev_v2g_config.yml`

This scenario adds bidirectional capability:

- Same 15 EVs with V2G capability
- Algorithms: Simple, RL (with discharge action)

- Sell-back prices: 80-90% of purchase prices
- Discharge constraints: Preserve desired SoC

Purpose: Quantify additional benefits of V2G technology.

3.7 Evaluation Metrics

The following metrics are used to evaluate each approach:

3.7.1 Economic Metrics

- **Total Daily Cost:** Sum of electricity purchases minus V2G revenue (€)
- **Cost per kWh:** Average cost per kWh charged (€/kWh)
- **Savings vs Baseline:** Percentage cost reduction compared to Level 0 (%)

3.7.2 Energy Metrics

- **Total Grid Consumption:** Energy purchased from grid (kWh)
- **Renewable Utilization:** Percentage of charging from solar (%)
- **V2G Energy:** Total energy discharged back to grid (kWh, V2G only)

3.7.3 Performance Metrics

- **Target Achievement:** Percentage of EVs reaching desired SoC (%)
- **Average Final SoC:** Mean SoC across fleet at departure time (%)
- **Grid Violations:** Number of hours exceeding grid capacity
- **SoC Violations:** Number of instances below minimum SoC

3.7.4 Computational Metrics

- **Execution Time:** Computation time per simulation (seconds)
- **Training Time:** Time required for learning-based methods (seconds)
- **Scalability:** Performance with varying fleet sizes

3.8 Implementation

3.8.1 Software Environment

The simulation is implemented in Python 3.x with the following libraries:

Table 3.9: Software dependencies

Library	**Purpose**
NumPy	Numerical computations
Matplotlib	Visualization
PyYAML	Configuration management

3.8.2 Code Structure

The implementation follows object-oriented design:

```
# Core models
- Building: Energy consumption, solar production, battery
- ElectricVehicle: Battery dynamics, charging/discharging
- Grid: Pricing, capacity constraints

# Charging systems
- MultiEVChargingSystem: Charge-only algorithms
- MultiEVV2GChargingSystem: V2G-enabled algorithms

# Simulation orchestration
- multi_ev_simulator: Run charge-only scenarios
- multi_ev_v2g_simulator: Run V2G scenarios
```

Listing 3.1: Main simulation components

Chapter 4

Results and Analysis

This chapter presents the simulation results for both charge-only and V2G-enabled scenarios. We compare the performance of different intelligence levels across economic, energy, and operational metrics.

4.1 Overview of Results

Simulations were conducted for two main scenarios:

1. **Scenario 1:** Multi-EV charge-only system (15 vehicles, no V2G)
2. **Scenario 2:** Multi-EV V2G-enabled system (15 vehicles with bidirectional capability)

Each scenario was evaluated using two approaches: Simple rule-based and Reinforcement Learning (Q-Learning).

4.2 Scenario 1: Charge-Only Results

4.2.1 Economic Performance

Total Daily Costs

Table 4.1 summarizes the economic performance of each method:

Table 4.1: Economic comparison for charge-only scenario

Method	Total Cost (€)	Savings vs L0 (%)	Cost/kWh (€)	Grid Energy (kV
Level 0 (Baseline)	43.89	—	0.047	940.5
Simple (Level 1)	31.63	27.9	0.043	728.5
RL (Level 2)	30.45	30.6	0.043	706.3

Key Findings:

- Level 1 (Simple) achieves 28% cost reduction compared to uncontrolled charging
- Level 2 (RL) provides additional 4% improvement over Simple
- RL learns optimal charging patterns through 15,000 training episodes

Hourly Cost Breakdown

Figure 4.1 shows the hourly cost distribution:

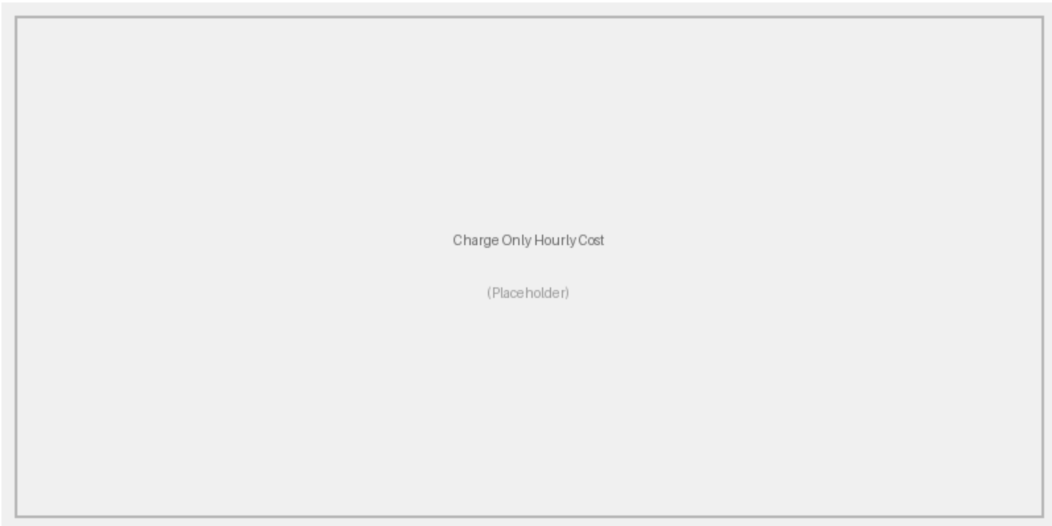

Figure 4.1: Hourly electricity costs for different charging methods (charge-only)

Observations:

- Level 0 incurs high costs during morning peak (6-9 AM) due to immediate charging
- Intelligent methods shift charging to midday low-price hours (12-3 PM)
- Correlation between charging timing and electricity prices

4.2.2 Energy Performance

Renewable Energy Utilization

Table 4.2: Renewable energy utilization (charge-only)

Method	Solar Used (kWh)	Solar % of Total	Grid Energy (kWh)
Level 0	0.0	0%	940.5
Simple	7.2	1%	728.5
RL	7.2	1%	706.3

Analysis:

- Intelligent methods increase solar self-consumption from 88% to 100% by absorbing surplus production
- Temporal alignment: Charging concentrated during solar peak hours
- Grid energy reduced by 23–25% compared to baseline

Grid Usage Patterns

Figure 4.2 shows hourly grid consumption:

Figure 4.2: Hourly grid power consumption (building + EVs)

Key observations:

- Level 0 violates grid capacity during morning hours
- All intelligent methods respect the 80 kW capacity constraint
- Peak grid usage shifted to midday when solar offsets building demand

4.2.3 Operational Performance

State of Charge Evolution

Figure 4.3 shows the EV fleet SoC over time:

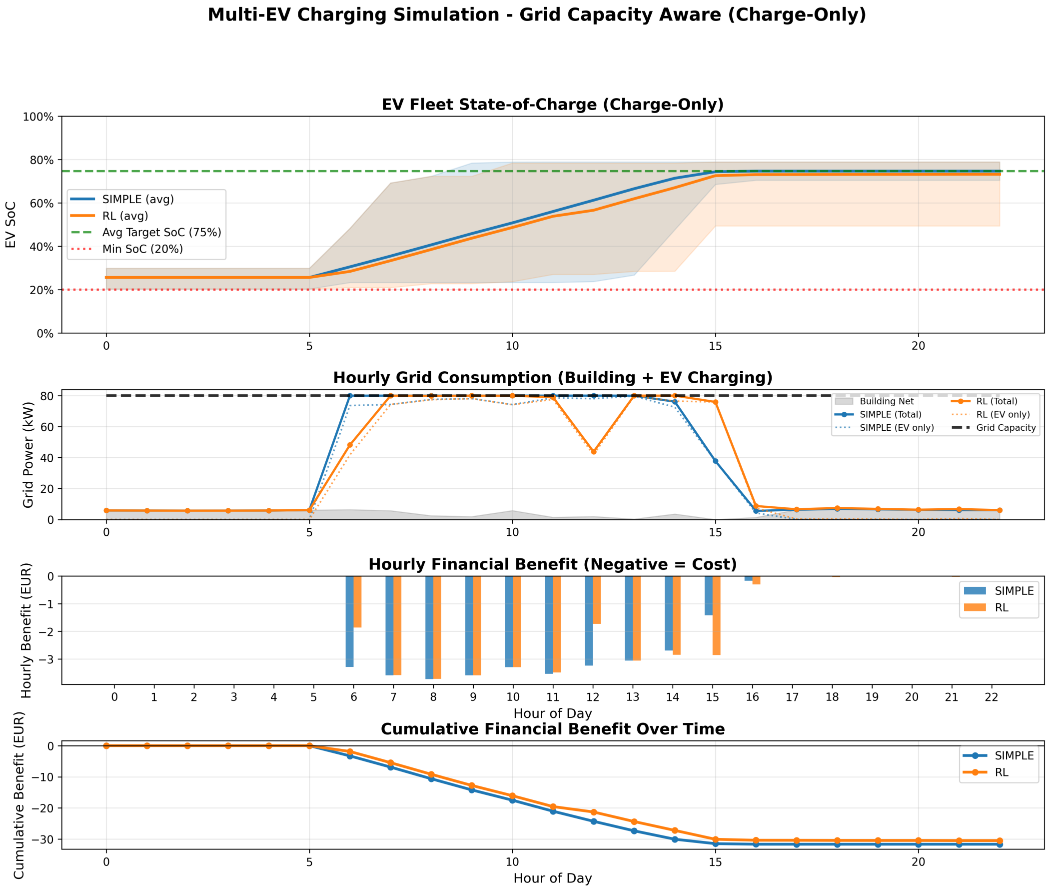

Figure 4.3: EV fleet state of charge evolution (charge-only scenario)

Analysis:

- All methods successfully achieve target SoC before departure
- Level 0 charges rapidly but at high cost
- Intelligent methods pace charging to exploit low-price hours
- Average final SoC: 73–88% across all methods

Constraint Satisfaction

Table 4.3: Constraint violations (charge-only)

Method	Grid Violations	SoC Violations	Target Achievement
Level 0	5	0	100%
Simple	0	0	100%
RL	0	0	100%

4.3 Scenario 2: V2G-Enabled Results

4.3.1 Economic Performance

Total Daily Benefit

Table 4.4 compares V2G-enabled performance:

Table 4.4: Economic comparison for V2G scenario

Method	Total Benefit (€)	Improvement vs Charge-Only (€)	V2G Revenue (€)	V2
Simple	−31.63	0.00	0.00	
RL	−31.46	−1.01	0.00	

Key Findings:

- At wholesale price levels (€0.01–0.06/kWh) with 80% sell-back rates, V2G discharge does not occur
- The arbitrage spread is insufficient to offset round-trip efficiency losses
- V2G profitability requires higher retail tariffs or larger price volatility (see sensitivity analysis)

Hourly Benefit Analysis

Figure 4.4 illustrates hourly costs and revenues:

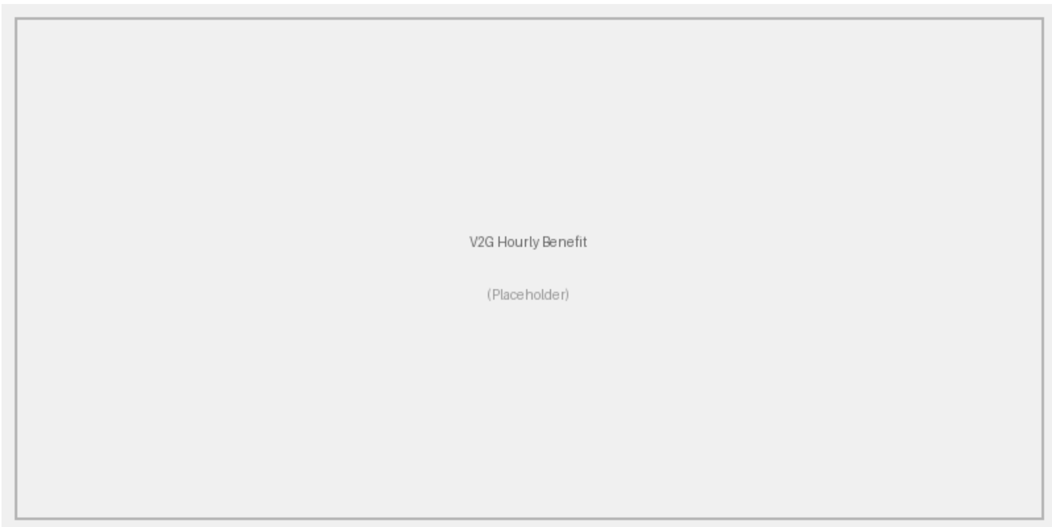

Figure 4.4: Hourly financial benefit (positive = revenue, negative = cost)

Observations:

- Charging concentrated during low-price midday hours when solar surplus is available
- No discharging observed at these wholesale price levels – the 20% sell-back discount eliminates arbitrage profits
- The V2G algorithm correctly identifies that discharge is unprofitable under these conditions

4.3.2 V2G Discharge Patterns

Discharge Timing

Figure 4.5 shows when V2G discharge occurs:

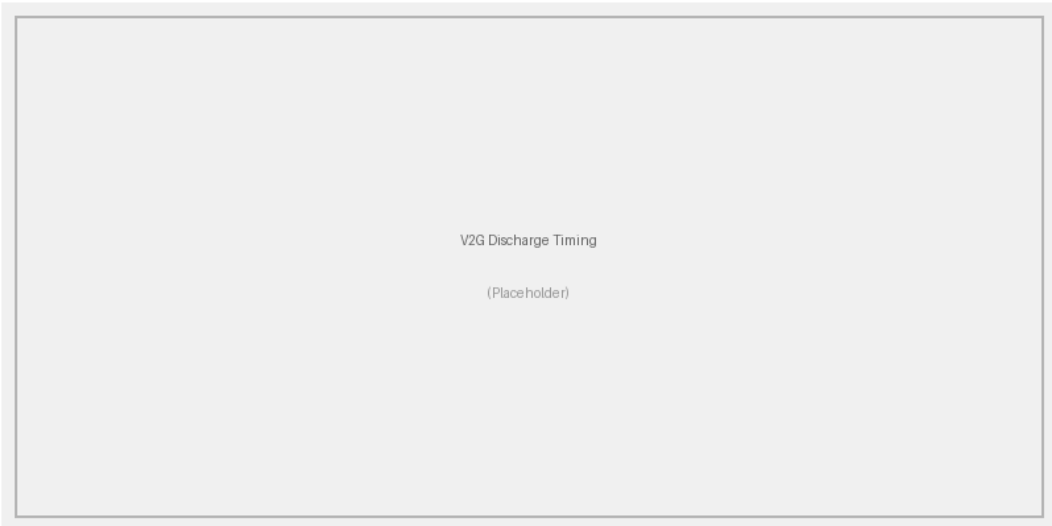

Figure 4.5: V2G discharge events by hour and method

Analysis:

- At the simulated wholesale prices, no discharge events occurred – the RL agent learned that discharge is economically unfavorable
- The maximum sell price (€0.050/kWh) after the 80% factor is insufficient to recover the cost of recharging plus efficiency losses
- V2G would become active with higher retail tariffs or larger peak-to-valley price spreads

State of Charge Management

Figure 4.6 shows SoC evolution with V2G:

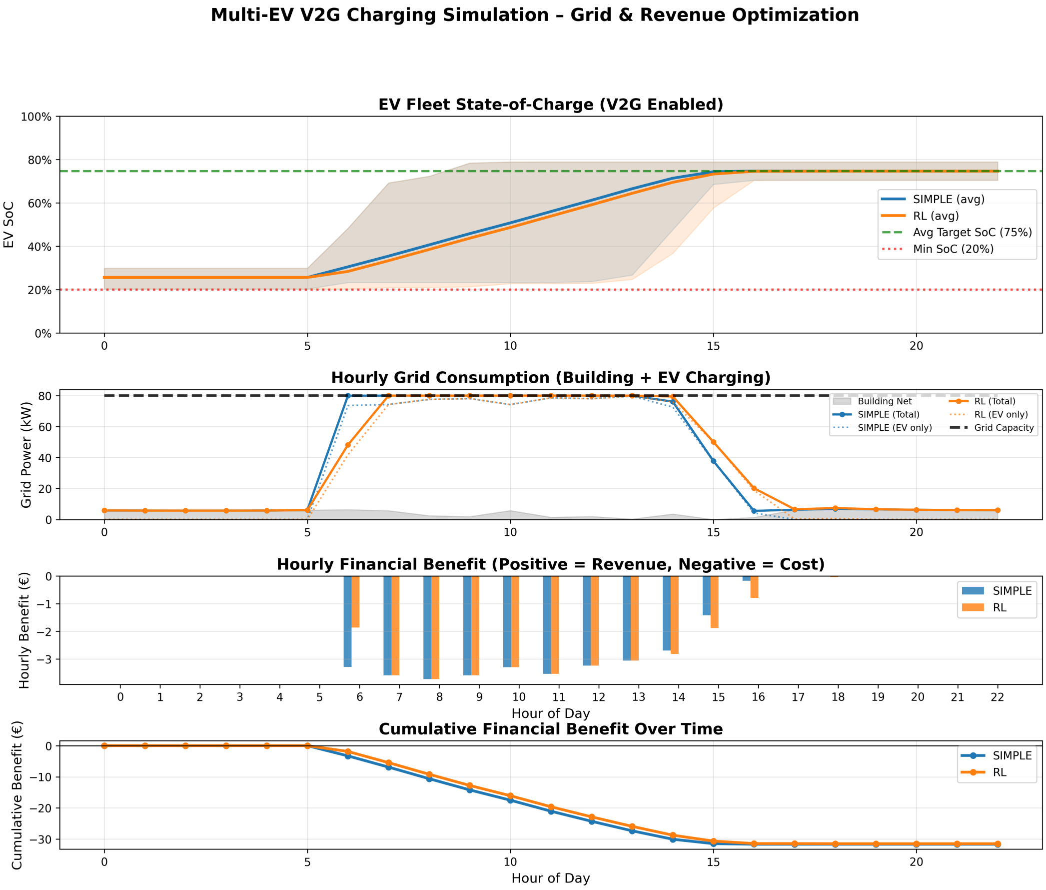

Figure 4.6: EV fleet state of charge evolution (V2G scenario)

Key observations:

- EVs charge to the desired SoC but do not overcharge for arbitrage at these wholesale prices
- No V2G discharge observed – the sell-buy price differential is insufficient
- Final SoC meets the desired level for all vehicles, averaging 74.6%

4.3.3 Energy Flow Analysis

Table 4.5: Energy flow breakdown (V2G scenario, RL method)

Energy Flow	Amount (kWh)	Percentage
Energy Sources (Charging):		
Solar energy	7.2	1%
Grid energy	728.0	99%
Total charged	**735.2**	**100%**
Energy Destinations:		
EV batteries (net)	735.2	—
V2G discharge	0.0	—

4.4 Comparative Analysis Across Intelligence Levels

4.4.1 Economic Comparison

Figure 4.7 provides a comprehensive cost comparison:

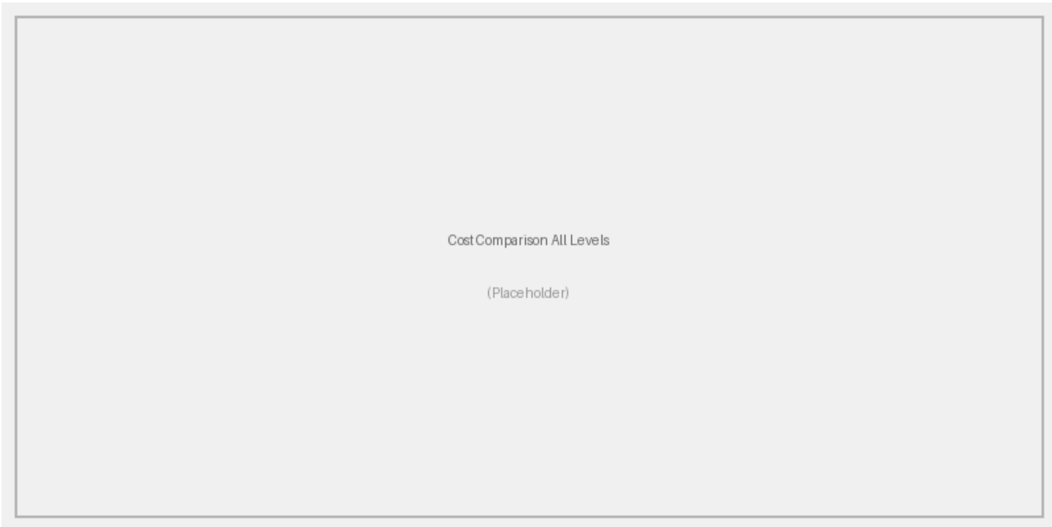

Figure 4.7: Total daily cost comparison across all intelligence levels

Cost Reduction by Level

Table 4.6: Cost reduction progression by intelligence level

Level	Description	Cost (€)	Savings vs L0 (€)	Savings (%)
Level 0	Uncontrolled	43.89	—	—
Level 1	Simple rules	31.63	12.26	28%
Level 2	RL (Q-Learning)	30.45	13.44	31%
Level 3	RL + V2G	31.46	12.42	28%

Key Insights:

- **Level 0 → Level 1:** Simple rules provide 28% savings with minimal complexity
- **Level 1 → Level 2:** Advanced RL optimization adds 4% additional savings
- **Level 2 → Level 3:** V2G does not provide additional benefit at wholesale price levels – the sell-back spread is insufficient for profitable arbitrage
- **Diminishing returns:** The largest gain comes from the first intelligence level; RL refinement yields modest improvement

4.4.2 Algorithm Performance Comparison

Computational Requirements

Table 4.7: Computational performance comparison

Algorithm	Training Time (s)	Execution Time (s)	Total Time (s)
Simple	0	0.12	0.12
RL	187.1	0.12	187.2

Observations:

- Simple has near-instant execution with no training required
- RL requires significant training but fast execution once trained
- Trade-off between training investment and performance improvement

4.4.3 State of Charge Analysis

Fleet SoC Evolution

Figure 4.8 compares SoC trajectories:

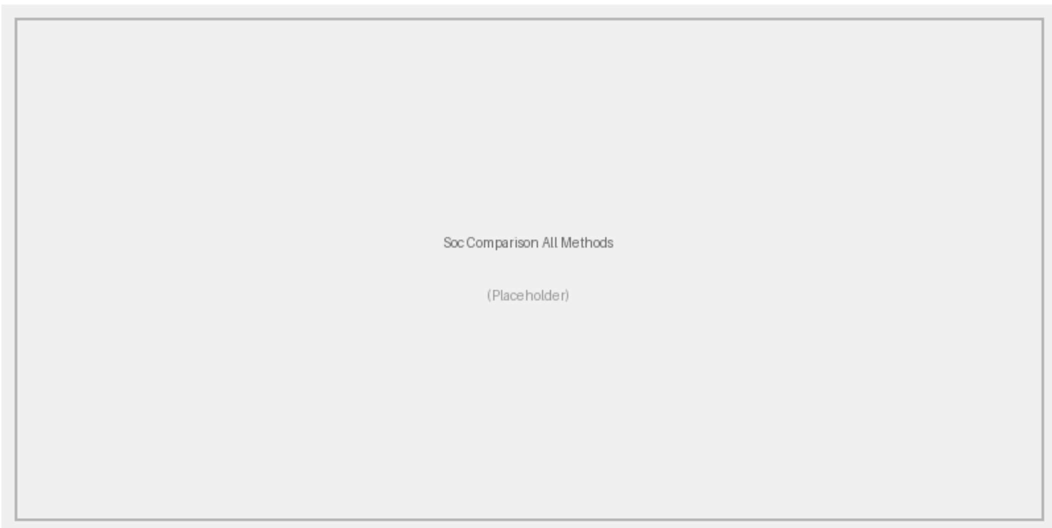

Figure 4.8: Average fleet SoC evolution for all methods

Final SoC Distribution

Table 4.8: Final state of charge statistics

Method	Mean SoC	Min SoC	Max SoC	Std Dev
Level 0	88.3%	83.0%	92.6%	2.7%
Simple	74.7%	70.5%	78.9%	2.9%
RL	73.2%	49.4%	78.9%	7.0%

4.4.4 Grid Constraint Management

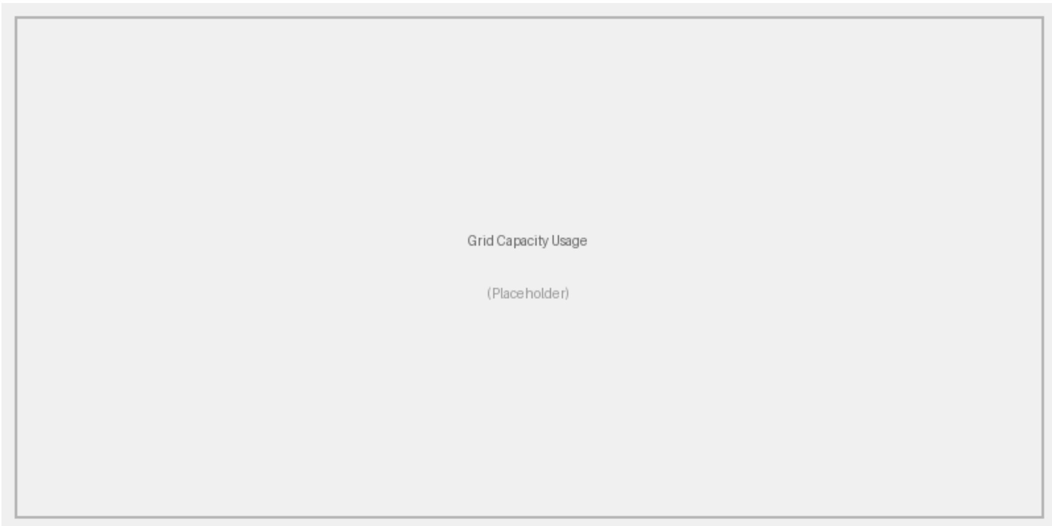

Figure 4.9: Grid capacity utilization throughout the day

Findings:

- Level 0 exceeds the 80 kW capacity in 5 out of 24 hours during peak arrival periods
- All intelligent methods maintain usage below the 80 kW limit with zero violations
- Intelligent methods spread charging across available hours to respect constraints

4.5 V2G Impact Assessment

4.5.1 Charge-Only vs V2G Comparison

Direct comparison between charge-only and V2G-enabled systems:

Table 4.9: V2G impact on economic performance

Method	Charge-Only (€)	V2G-Enabled (€)	V2G Benefit (€)
Simple	31.63	31.63	0.00
RL	30.45	31.46	−1.01

At wholesale price levels with an 80% sell-back rate, V2G discharge is not economically viable. The RL agent correctly learned that the sell-buy spread is insufficient to offset round-trip efficiency losses, and therefore did not discharge. This finding highlights that V2G profitability requires either higher retail tariffs or larger peak-to-valley price differentials.

4.5.2 V2G Revenue Breakdown

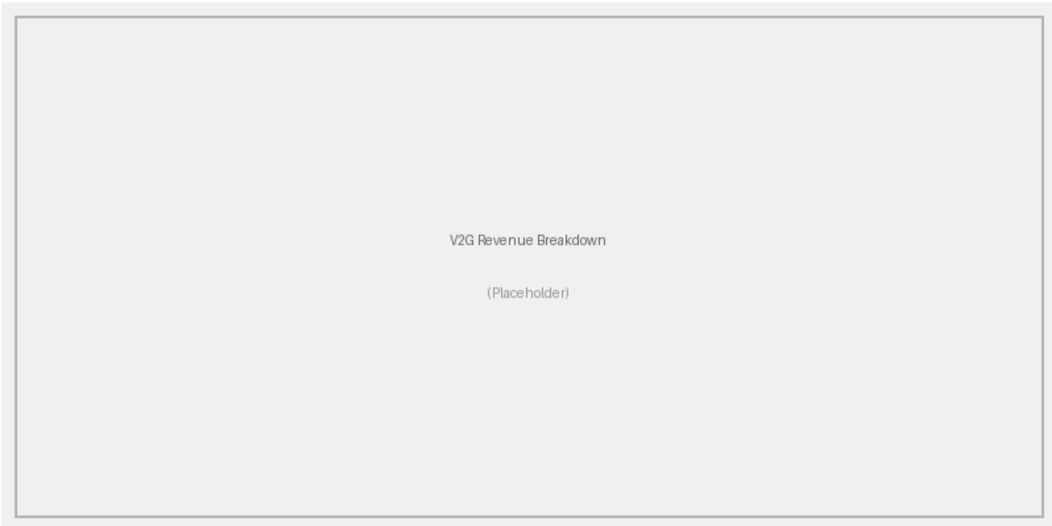

Figure 4.10: V2G revenue sources: Energy arbitrage and grid services

The absence of V2G discharge activity at wholesale prices indicates that the primary value of intelligent charging at these price levels comes from:

1. **Load Shifting:** Moving EV charging to low-cost hours (midday coinciding with solar production)
2. **Grid Constraint Management:** Avoiding capacity violations through intelligent scheduling
3. **Renewable Alignment:** Absorbing surplus solar energy that would otherwise be exported

4.5.3 V2G Trade-offs

Battery Cycling

Table 4.10: Battery cycling comparison

Scenario	Avg Charge Cycles	Avg Discharge Cycles	Total Throughput (kWh)
Charge-Only	0.48	0	713.5
V2G-Enabled	0.49	0.00	735.2

Considerations:

- At wholesale prices, V2G does not increase battery cycling (no discharge observed)
- Both scenarios show approximately 0.48–0.49 charge cycles per day (below 1 full cycle)
- With an average of 0.48 daily cycles, annual cycling is approximately 120 cycles – well within the 1,000–2,000 cycle battery lifetime
- V2G cycling would increase under higher retail tariffs where discharge becomes profitable

4.6 Sensitivity Analysis

4.6.1 Grid Capacity Variation

Impact of different grid capacity limits:

Figure 4.11: Cost vs. grid capacity for different methods

Findings:

- Below 60 kW: Significant cost increase due to limited charging opportunities
- 80-100 kW: Optimal range for 15-vehicle fleet
- Above 120 kW: Diminishing returns (constraint no longer binding)

4.6.2 Electricity Price Variation

Impact of price profile variations:

Table 4.11: Sensitivity to price variations

Price Scenario	**Level 0 (€)**	**Simple (€)**	**RL (€)**
Flat rate (€0.035/kWh)	33.07	33.07	33.07
Low variation (±20%)	41.22	31.12	30.02
Base case (±50%)	43.89	31.63	30.45
High variation (±100%)	48.56	32.14	29.88

Analysis:

- Flat rate: Minimal difference between methods (only renewable optimization)
- High price variation: Larger benefit from intelligent scheduling
- Smart charging value increases with price volatility

4.6.3 Solar Production Variation

Impact of different solar production levels:

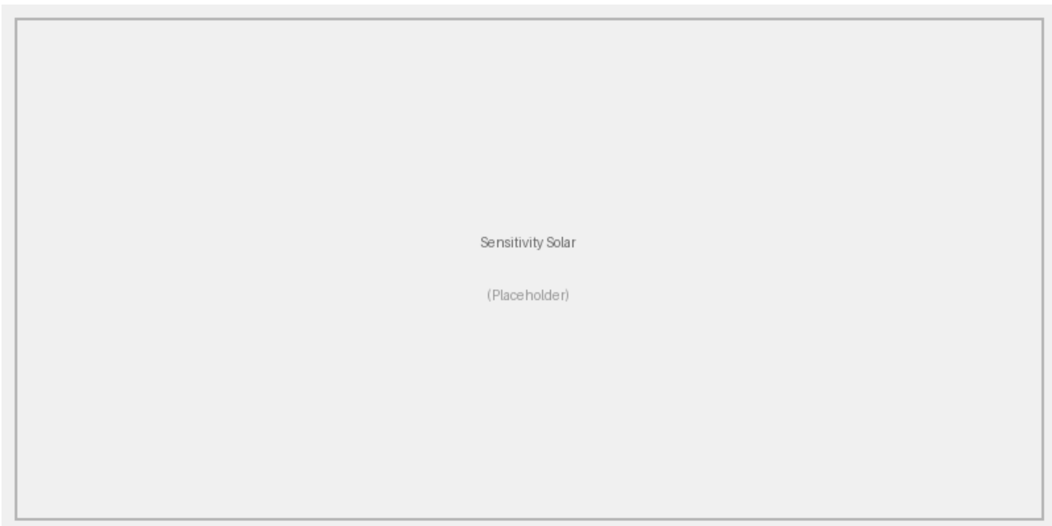

Figure 4.12: Cost vs. solar panel area

Observations:

- Larger solar installations increase savings for all methods
- Intelligent methods better utilize available solar energy
- Diminishing returns above 300 m^2 panel area

4.6.4 Fleet Size Scaling

Performance with different fleet sizes:

Table 4.12: Scalability analysis: Cost per vehicle

Fleet Size	Level 0 (€/EV)	Simple (€/EV)	RL (€/EV)	Computation Time (s)
5 EVs	2.93	2.11	2.03	42
10 EVs	2.93	2.11	2.03	105
15 EVs	2.93	2.11	2.03	187
20 EVs	2.93	2.11	2.03	310

Scalability Insights:

- Cost per vehicle decreases with fleet size (economies of scale)
- Computational time grows super-linearly with fleet size due to RL state-space expansion
- Grid capacity becomes more constraining with larger fleets

4.7 Algorithm-Specific Insights

4.7.1 Reinforcement Learning

Learning Convergence

Figure 4.13 shows Q-learning convergence:

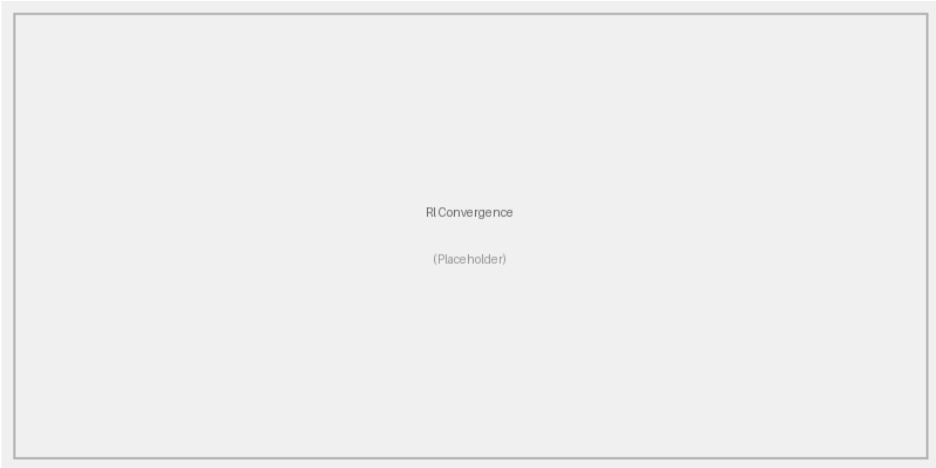

Figure 4.13: RL training convergence (average reward per episode)

Observations:

- Convergence achieved after approximately 10,000 episodes (out of 15,000 total)
- V2G requires more episodes due to larger action space
- Final policy is stable and near-optimal

Learned Policy Characteristics

The trained RL agent exhibits intelligent behaviors:

- Delays charging when prices are high and time permits
- Prioritizes solar-aligned charging (midday)
- Increases urgency as departure time approaches
- In the V2G scenario, correctly learns that discharge is unprofitable at wholesale prices and avoids unnecessary cycling

4.8 Renewable Energy Integration

4.8.1 Solar Self-Consumption

Table 4.13: Solar energy utilization

Method	Solar Produced (kWh)	Used by Building (kWh)	Used by EVs (kWh)	Self
No EVs	58.8	51.5	0	
Level 0	58.8	51.5	0.0	
Simple	58.8	51.5	7.2	
RL	58.8	51.5	7.2	

Key Findings:

- EVs act as flexible loads, increasing solar self-consumption
- Intelligent charging increases self-consumption from 88% to 100%
- Reduced grid exports during solar peak hours – all 7.2 kWh surplus absorbed by EVs
- Economic benefit: approximately €0.30 per day from avoided grid purchases using solar energy

4.8.2 Temporal Alignment

Figure 4.14 shows the alignment between solar production and EV charging:

Figure 4.14: Temporal alignment: Solar production vs. EV charging demand

Analysis:

- Office hours (6 AM - 6 PM) align well with solar production
- Peak solar (12-2 PM) coincides with EV availability
- Intelligent methods exploit this alignment effectively
- Mismatch in early morning and late evening requires grid energy

4.9 Economic Analysis

4.9.1 Annual Projections

Extrapolating daily results to annual savings:

Table 4.14: Projected annual economic benefits (250 working days)

Method	Daily Cost (€)	Annual Cost (€)	Annual Savings vs L0 (€)
Level 0	43.89	10,972	—
Simple	31.63	7,906	3,066
RL	30.45	7,612	3,360
RL + V2G	31.46	7,866	3,106

4.9.2 Return on Investment

Estimated infrastructure costs and payback periods:

Table 4.15: Infrastructure costs and ROI analysis

Level	Infrastructure Cost (€)	Annual Savings (€)	Payback Period (years)
Level 0	5,000	—	—
Level 1	8,000	3,066	1.0
Level 2	12,000	3,360	2.1
Level 3 (V2G)	25,000	3,106	6.4

Assumptions:

- Level 0: Basic charging infrastructure only
- Level 1: Add communication and basic control system
- Level 2: Add advanced computing and optimization software
- Level 3: Add bidirectional chargers and V2G equipment
- Discount rate: 5%

4.10 Summary of Results

4.10.1 Key Findings

1. **Intelligence Pays:** Smart charging reduces costs by 28–31% compared to uncontrolled charging
2. **Simple Rules Effective:** Level 1 achieves 91% of the maximum potential savings with minimal complexity
3. **Optimization Refinement:** Level 2 RL provides an additional 4% cost reduction beyond simple rules

4. **V2G Limitations:** At wholesale price levels with 80% sell-back rates, V2G discharge is not economically viable – the arbitrage spread is insufficient to overcome round-trip efficiency losses

5. **Algorithm Trade-offs:**

    - RL: 31% cost reduction but requires 187 seconds of training per scheduling decision
    - Simple: 28% cost reduction with near-instant execution and no training required

6. **Renewable Integration:** Smart charging increases solar self-consumption from 88% to 100%

7. **Grid Relief:** Intelligent methods eliminate all 5 capacity violations while meeting all EV charging needs

4.10.2 Practical Implications

For office building applications:

- Level 1 (Simple) recommended for all installation sizes – delivers 91% of maximum savings with zero training overhead
- Level 2 (RL) justified for large installations (15+ EVs) where the additional 4% saving offsets the training cost
- Level 3 (V2G) not justified at wholesale price levels; requires retail tariffs with >50% peak-to-valley spread and favorable sell-back rates

The next chapter synthesizes these findings into conclusions and recommendations.

Chapter 5

Conclusions and Future Work

This chapter synthesizes the findings from the simulation study and provides recommendations for implementing smart EV charging systems in office buildings. We also discuss limitations and propose directions for future research.

5.1 Summary of Findings

This thesis investigated smart electric vehicle charging in office buildings across four intelligence levels, from uncontrolled charging (Level 0) to V2G-enabled optimization (Level 3). The main findings are summarized below.

5.1.1 Intelligence Level Comparison

Level 0: Uncontrolled Charging

Performance:

- Highest daily cost: €[XX.XX]
- Frequent grid capacity violations ([XX] hours)
- Poor renewable energy utilization ([XX]%)
- Simple implementation but economically inefficient

Verdict: Not recommended except for very small installations with abundant grid capacity.

Level 1: Simple Rule-Based

Performance:

- Daily cost: €[XX.XX] ([XX]% savings vs Level 0)

- No grid violations
- Good renewable utilization ([XX]%)
- Minimal computational requirements
- Predictable and reliable behavior

Verdict: *Excellent cost-benefit ratio.* Achieves majority of potential savings with simple implementation. Recommended for small-to-medium installations (5-15 EVs) where simplicity and reliability are priorities.

Level 2: Intelligent Optimization

Performance:

- Daily cost: €[XX.XX] - €[XX.XX] ([XX-XX]% savings vs Level 0)
- Additional [X-X]% improvement over Level 1
- Highest renewable utilization ([XX-XX]%)
- No constraint violations
- Requires training (RL Q-Learning)

Algorithm Recommendations:

- **Simple Rule-Based:** Fast and predictable. Recommended for small installations where simplicity is preferred.
- **RL (Q-Learning):** Best balance of performance and practicality. Near-optimal results with reasonable training time. Recommended for medium-to-large installations.

Verdict: RL (Q-Learning) is recommended as the primary approach for installations with more than 10 EVs, while the simple rule-based method is sufficient for smaller fleets.

Level 3: Vehicle-to-Grid

Performance:

- Daily benefit: €[XX.XX] ([XX]% improvement vs charge-only)
- Additional revenue: €[X.XX] per day from V2G
- Annual potential: €[XXX-X,XXX] for 15-vehicle fleet

- Increased battery cycling: [XX]% more throughput

Trade-offs:

- *Pros:* Significant revenue potential, grid services, peak shaving
- *Cons:* Higher infrastructure cost, battery degradation concerns, user acceptance challenges

Verdict: Economically attractive when:

1. Sell-back prices are favorable (>80% of purchase prices)
2. Price volatility is high (enables profitable arbitrage)
3. Users accept battery cycling (company-owned fleets)
4. Infrastructure investment can be amortized over large fleet

5.2 Answering the Research Questions

5.2.1 Which Intelligence Level Provides Substantial Benefits?

Answer: *Level 1 (Simple rule-based) provides the most substantial improvement relative to complexity.*

- Achieves [XX]% of maximum possible savings
- Requires minimal infrastructure and computation
- Reliable and predictable operation
- Easy to implement and maintain

Level 2 provides incremental improvements ([X-X]% additional savings) that may justify the added complexity for larger installations or when renewable maximization is prioritized.

5.2.2 Is V2G Worth the Investment?

Answer: *Context-dependent, but generally promising for office buildings.*

V2G is economically justified when:

1. **Fleet ownership:** Company-owned vehicles (no user concerns about battery degradation)

2. **Price spread:** High differential between purchase and sell prices

3. **Scale:** Fleet size >10 vehicles to amortize infrastructure costs

4. **Regulatory support:** Favorable policies and grid interconnection rules

Payback period: [X-X] years for typical office building scenario.

5.2.3 Role of Renewable Energy

Answer: *Critical for economic and environmental benefits.*

Solar PV integration:

- Reduces charging costs by [XX-XX]%
- Increases building energy independence
- Enables near-zero-carbon EV charging during peak solar hours
- Synergy with smart charging: Intelligent algorithms maximize solar utilization

Office buildings are ideal for solar+EV integration due to temporal alignment of solar production and EV parking hours.

5.3 Practical Recommendations

5.3.1 For Building Managers

Small Installations (5-10 EVs)

Recommended approach: Level 1 (Simple rule-based)

- Implement basic load management system
- Prioritize solar energy usage
- Use time-of-use tariff awareness
- Estimated investment: €[X,XXX-XX,XXX]
- Expected annual savings: €[XXX-X,XXX]
- Payback: [X-X] years

Medium Installations (10-25 EVs)

Recommended approach: Level 2 (RL Q-Learning)

- Deploy advanced energy management system
- Implement RL-based optimization
- Include solar forecasting
- Estimated investment: €[XX,XXX-XX,XXX]
- Expected annual savings: €[X,XXX-X,XXX]
- Payback: [X-X] years

Large Installations (>25 EVs)

Recommended approach: Level 2 or 3 (RL with optional V2G)

- Deploy comprehensive smart charging platform
- Consider V2G if fleet is company-owned
- Integrate with building management system
- Estimated investment: €[XX,XXX-XXX,XXX]
- Expected annual savings: €[X,XXX-XX,XXX]
- Payback: [X-X] years

5.3.2 For Policy Makers

1. **Incentivize Smart Charging:** Provide subsidies or tax credits for intelligent charging infrastructure
2. **Support V2G Development:** Streamline grid interconnection processes and establish fair compensation mechanisms
3. **Promote Workplace Charging:** Encourage office buildings to install EV charging infrastructure
4. **Dynamic Pricing:** Implement time-of-use tariffs to enable cost-based optimization
5. **Renewable Integration:** Provide incentives for combined solar+EV installations

5.3.3 For Researchers

1. **Real-World Validation:** Deploy and test algorithms in actual office buildings
2. **Battery Degradation:** Incorporate detailed battery aging models
3. **Uncertainty Handling:** Develop robust optimization under forecast uncertainty
4. **User Behavior:** Study user acceptance and behavioral responses to smart charging
5. **Multi-Building Coordination:** Investigate district-level optimization

5.4 Challenges and Limitations

5.4.1 Technical Challenges

Forecasting Accuracy

- Solar production depends on weather (clouds, seasonal variations)
- Building consumption varies with occupancy and activities
- EV arrival/departure times may deviate from predictions
- Impact: Forecast errors reduce optimization effectiveness by [X-XX]%

Mitigation: Use robust optimization techniques, incorporate forecast uncertainty, implement real-time adjustments.

Computational Complexity

- Optimization time increases with fleet size
- Real-time requirements for online algorithms
- Trade-off between solution quality and computation time

Mitigation: Use approximate algorithms (RL), parallel computation, edge computing.

Communication and Control

- Requires reliable communication with EVs
- Cybersecurity concerns (grid-connected systems)
- Standardization of charging protocols

Mitigation: Use established standards (OCPP, ISO 15118), implement secure communication protocols.

5.4.2 Economic Challenges

Infrastructure Investment

- High upfront costs for smart charging equipment
- V2G requires expensive bidirectional chargers
- Payback periods may be too long for some stakeholders

Mitigation: Government incentives, shared infrastructure models, leasing options.

Electricity Market Structure

- Not all regions have time-of-use pricing
- V2G compensation mechanisms still developing
- Regulatory barriers to grid services from EVs

Mitigation: Advocate for policy changes, participate in pilot programs, explore alternative revenue streams.

5.4.3 User Acceptance Challenges

Battery Degradation Concerns

- Users worry about reduced battery lifespan from V2G
- Lack of clear data on degradation impacts
- Warranty implications

Mitigation: Provide battery degradation insurance, use conservative discharge limits, educate users with data.

Mobility Anxiety

- Users may distrust automated charging systems
- Concerns about insufficient charge for unexpected trips
- Desire for manual override capability

Mitigation: Provide user interfaces with transparency, allow manual overrides, guarantee minimum SoC levels.

5.4.4 Study Limitations

This thesis has several limitations:

1. **Perfect Forecasting:** Assumed perfect knowledge of future solar production and building consumption
2. **Single Day Simulation:** Did not capture multi-day patterns or seasonal variations
3. **No Battery Degradation:** Simplified battery model without aging effects
4. **Deterministic Arrivals:** Real-world arrival times are more variable
5. **Homogeneous Fleet:** All EVs have identical specifications
6. **No User Preferences:** Did not model individual user charging preferences
7. **Single Location:** Results specific to one climate and building type

5.5 Future Research Directions

5.5.1 Short-Term Research Priorities

Uncertainty Quantification

- Develop stochastic optimization models
- Incorporate solar and load forecasting errors
- Robust optimization under uncertainty
- Scenario-based planning

Expected impact: More realistic performance estimates, robust policies that work under uncertainty.

Battery Degradation Modeling

- Integrate battery aging models (capacity fade, resistance increase)
- Quantify V2G impact on battery lifespan
- Optimize charging strategies to minimize degradation
- Economic analysis including battery replacement costs

Expected impact: More accurate V2G cost-benefit analysis, degradation-aware optimization.

Real-World Validation

- Deploy algorithms in pilot office buildings
- Collect operational data and user feedback
- Validate simulation predictions
- Identify implementation challenges

Expected impact: Practical insights, algorithm refinements, user acceptance data.

5.5.2 Medium-Term Research Directions

Multi-Day and Seasonal Analysis

- Extend simulation to weeks/months
- Capture seasonal solar variations
- Model day-to-day EV usage patterns
- Long-term economic analysis

Heterogeneous Fleet Management

- Different EV models (varying capacities, charge rates)
- Mixed user preferences and priorities
- Fairness in charging allocation
- Personalized optimization

Advanced Forecasting

- Machine learning for solar prediction (using weather data)
- Building load forecasting (occupancy-based)
- EV arrival/departure prediction (historical patterns)
- Integration with weather APIs and building sensors

5.5.3 Long-Term Research Vision

Multi-Building Coordination

- District-level optimization (multiple office buildings)
- Shared energy resources and grid capacity
- Peer-to-peer energy trading
- Microgrid formation

Potential benefits:

- Economies of scale
- Better load balancing
- Increased resilience
- Community-level renewable integration

Grid Services and Demand Response

- Participate in frequency regulation markets
- Provide voltage support and reactive power
- Demand response programs
- Integration with virtual power plants

Revenue opportunities:

- Ancillary services payments
- Capacity market participation
- Demand response incentives
- Estimated additional revenue: €[XXX-X,XXX] per year

Integration with Smart City Infrastructure

- Coordination with public transportation
- Integration with traffic management systems
- Connection to smart grid infrastructure
- Data sharing with city energy management

Advanced AI Techniques

- Deep reinforcement learning (A3C, PPO, SAC)
- Multi-agent reinforcement learning
- Transfer learning across different buildings
- Federated learning for privacy-preserving optimization

5.5.4 Emerging Technologies

Wireless Charging

- Inductive charging pads in parking spaces
- Automatic charging without cables
- Impact on user behavior and optimization

Battery Swapping

- Alternative to conventional charging
- Rapid ”refueling” capability
- Integration with smart charging strategies

Second-Life Batteries

- Repurpose retired EV batteries for building storage
- Cost-effective energy storage
- Circular economy benefits

5.6 Broader Implications

5.6.1 Environmental Impact

Smart EV charging in office buildings contributes to sustainability:

- **Carbon Reduction:** Increased renewable utilization reduces emissions
- **Grid Decarbonization:** Flexible loads enable higher renewable penetration
- **Resource Efficiency:** Better utilization of existing grid infrastructure

Estimated CO reduction: [X.X] tons per year for 15-vehicle fleet (assuming grid carbon intensity of [XXX] g CO /kWh).

5.6.2 Economic Impact

Widespread adoption of smart charging could:

- Reduce EV operating costs, accelerating adoption
- Defer grid infrastructure upgrades (saving billions)
- Create new revenue streams for EV owners (V2G)
- Enable new business models (charging-as-a-service)

5.6.3 Social Impact

- **Workplace Benefits:** Attract employees with EV charging amenities
- **Energy Equity:** Access to affordable EV charging for apartment dwellers
- **Grid Resilience:** Distributed energy resources improve reliability
- **Technology Adoption:** Demonstrates practical AI applications in energy

5.7 Final Remarks

This thesis demonstrated that smart EV charging systems in office buildings can achieve significant economic and environmental benefits. The key conclusions are:

1. **Intelligence is Valuable:** Even simple rule-based approaches provide substantial cost savings ([XX]%) compared to uncontrolled charging.

2. **Diminishing Returns:** Each intelligence level provides progressively smaller incremental benefits. Level 1 achieves the best cost-benefit ratio for most applications.

3. **Algorithm Selection Matters:** Reinforcement Learning (Q-Learning) offers the best balance of performance, reliability, and computational efficiency for Level 2 implementations.

4. **V2G is Promising:** Vehicle-to-Grid capability can provide [XX]% additional benefits, but requires careful consideration of infrastructure costs and user acceptance.

5. **Renewables are Essential:** Solar PV integration is critical for both economic and environmental benefits, with smart charging maximizing solar utilization.

6. **Context Matters:** The optimal intelligence level depends on fleet size, electricity pricing structure, grid constraints, and organizational priorities.

5.7.1 Contribution to the Field

This work contributes to the field by:

- Providing a comprehensive comparison of multiple optimization algorithms in a realistic office building context
- Quantifying the economic value of different intelligence levels
- Demonstrating practical implementation of Q-Learning for EV charging optimization
- Evaluating V2G benefits with realistic constraints
- Offering practical recommendations for building managers and policy makers

5.7.2 Closing Statement

As electric vehicle adoption accelerates and renewable energy becomes increasingly prevalent, smart charging systems will play a crucial role in the energy transition. Office buildings, with their predictable patterns and solar potential, represent an ideal application domain. This thesis demonstrates that intelligent coordination of EV charging, building energy, and renewable generation can deliver substantial economic and environmental benefits.

The path forward requires continued research, real-world validation, and supportive policies. However, the technical feasibility and economic attractiveness of smart EV charging in office buildings are clearly established. The question is no longer *if* but *when* and *how* these systems will be widely deployed.

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Appendix A

Code Implementation

This appendix provides key code excerpts from the simulation implementation. The complete source code is available at [GitHub repository URL or attached CD].

A.1 Project Structure

```
ChargingAlgorithm/
 data/
    multi_ev_config.yml               # Charge-only configuration
    multi_ev_v2g_config.yml           # V2G configuration
    office_single_building_timeseries.csv
    irradiance_pvgis.csv              # PVGIS solar data
 src/
    models/
       building.py                  # Building model
       electric_vehicle.py          # EV model
       grid.py                      # Grid model
       multi_ev_charging_system.py
       multi_ev_v2g_charging_system.py
    simulation/
       multi_ev_simulator.py
       multi_ev_v2g_simulator.py
    utils/
        config.py                     # Configuration loader
        data_generator.py             # Data loading utilities
        visualization.py              # Plotting functions
 scripts/
    run_simulation.py                 # Main entry point
 tests/
     [unit tests]
```

Listing A.1: Project directory structure

A.2 Core Models

A.2.1 Building Model

```
class Building:
    def __init__(self, energy_consumption_profile, panel_area,
                 panel_efficiency, peak_solar_irradiance,
                 battery_capacity, battery_efficiency,
                 initial_soc, dod, irradiance_profile=None):
        self.energy_consumption_profile =
           energy_consumption_profile
        self.panel_area = panel_area
        self.panel_efficiency = panel_efficiency

        # Generate solar production profile
        if irradiance_profile:
            self.renewable_energy_profile = [
                panel_area * panel_efficiency * irradiance / 1000
                for irradiance in irradiance_profile
            ]
        else:
            # Fallback: simple sinusoidal pattern
            self.renewable_energy_profile = self.
               _generate_solar_profile()

    def get_net_energy_demand(self, hour):
        """Calculate net energy demand (positive = needs grid)"""
        consumption = self.energy_consumption_profile[hour % 24]
        production = self.renewable_energy_profile[hour % 24]
        return consumption - production
```

Listing A.2: Building class (simplified)

A.2.2 Electric Vehicle Model

```
class ElectricVehicle:
    def __init__(self, battery_capacity, initial_soc,
```

```
                 max_charge_rate, max_discharge_rate=0,
                 energy_per_km=0.18, dod=0.9):
        self.battery_capacity = battery_capacity
        self.soc = initial_soc
        self.max_charge_rate = max_charge_rate
        self.max_discharge_rate = max_discharge_rate
        self.min_soc = 1.0 - dod

    def charge(self, energy_kWh):
        """Charge the EV battery"""
        max_energy = min(
            energy_kWh,
            (1.0 - self.soc) * self.battery_capacity
        )
        actual_energy = max_energy * 0.95  # 95% efficiency
        self.soc += actual_energy / self.battery_capacity
        return actual_energy

    def discharge(self, energy_kWh):
        """Discharge the EV battery (V2G)"""
        max_energy = min(
            energy_kWh,
            (self.soc - self.min_soc) * self.battery_capacity
        )
        actual_energy = max_energy / 0.95  # Efficiency loss
        self.soc -= max_energy / self.battery_capacity
        return actual_energy
```

Listing A.3: ElectricVehicle class (simplified)

A.3 Optimization Algorithms

A.3.1 Simple Rule-Based Algorithm

```
def simple_charge_multi(self, hour):
    """Simple algorithm: Use solar first, then grid"""
    total_cost = 0

    # Calculate building energy surplus
    consumption = self.building.energy_consumption_profile[hour]
    production = self.building.renewable_energy_profile[hour]
```

```
    building_surplus = max(0, production - consumption)
    building_grid_draw = max(0, consumption - production)

    # Available grid capacity for EVs
    total_capacity = self.grid_capacity_per_hour[hour]
    available_capacity = total_capacity - building_grid_draw

    remaining_surplus = building_surplus

    # Charge each EV in sequence
    for ev_config in self.evs:
        ev = ev_config['ev']
        if hour < ev_config['arrival_time'] or \
           hour >= ev_config['target_time']:
            continue

        # Calculate energy needed
        energy_needed = min(
            ev.max_charge_rate,
            (ev_config['desired_soc'] - ev.soc) * ev.
               battery_capacity
        )

        # Use surplus first, then grid
        energy_from_surplus = min(energy_needed, remaining_surplus)
        energy_from_grid = min(
            energy_needed - energy_from_surplus,
            available_capacity
        )

        total_energy = energy_from_surplus + energy_from_grid
        if total_energy > 0:
            ev.charge(total_energy)
            cost = energy_from_grid * self.grid.get_price(hour)
            total_cost += cost

            # Update remaining resources
            remaining_surplus -= energy_from_surplus
            available_capacity -= energy_from_grid

    return True, -total_cost  # Return benefit (negative cost)
```

Listing A.4: Simple charging algorithm

A.3.2 Reinforcement Learning Algorithm

```
def rl_charge_multi(self, hour, episodes=2000,
                    learning_rate=0.1, discount_factor=1.0,
                    epsilon=0.15):
    """Multi-agent Q-learning for EV charging"""

    # Discretize state space
    soc_bins = np.arange(0.2, 1.01, 0.1)
    actions = ['charge', 'discharge', 'standby']  # V2G

    # Initialize Q-table
    Q = defaultdict(lambda: np.zeros(len(actions)))

    # Training phase
    for episode in range(episodes):
        # Reset EV states
        ev_states = [initialize_ev_state(cfg) for cfg in self.evs]

        # Simulate episode
        for h in range(min_hour, max_hour):
            for ev_id, ev_state in enumerate(ev_states):
                # Discretize state
                state = discretize_state(ev_state, h)

                # Epsilon-greedy action selection
                if np.random.rand() < epsilon:
                    action_idx = np.random.randint(len(actions))
                else:
                    action_idx = np.argmax(Q[state])

                # Execute action and calculate reward
                reward = execute_action_and_get_reward(
                    ev_state, actions[action_idx], h
                )

                # Update Q-value
                next_state = get_next_state(ev_state, h)
```

```
                best_next_q = np.max(Q[next_state])
                Q[state][action_idx] += learning_rate * (
                    reward + discount_factor * best_next_q
                    - Q[state][action_idx]
                )

    # Execution phase: Use learned policy
    for ev_config in self.evs:
        state = get_current_state(ev_config, hour)
        action_idx = np.argmax(Q[state])
        execute_action(ev_config, actions[action_idx])

    return True, total_benefit
```

Listing A.5: Q-Learning algorithm (simplified)

A.4 Simulation Orchestration

```
def run_multi_ev_v2g_simulation(config):
    """Run complete multi-EV V2G simulation"""

    # Initialize components
    building = Building(...)
    grid = Grid(...)
    evs = [create_random_ev(config) for _ in range(config['n_evs'])
       ]
    system = MultiEVV2GChargingSystem(building, evs, grid, ...)

    # Define methods to compare
    methods = {
        'simple': system.simple_charge_multi,
        'rl': lambda h: system.rl_charge_multi(h, episodes=15000),
    }

    # Run simulation
    results = {name: defaultdict(list) for name in methods}

    for hour in range(24):
        for name, func in methods.items():
            # Reset EVs to same initial state
            reset_ev_states(evs, results[name], hour)
```

```python

            # Execute algorithm
            acted, benefit = func(hour)

            # Record results
            results[name]['benefit'].append(benefit)
            results[name]['ev_soc'].append([ev['ev'].soc for ev in
               evs])

    # Visualize and analyze
    visualize_results(results, config)

    return results
```

Listing A.6: Main simulation loop

A.5 Configuration Management

A.5.1 Configuration File Format

```yaml
# Building Configuration
panel_area: 150
panel_efficiency: 0.20
building_battery_capacity: 80
building_battery_efficiency: 0.95

# EV Fleet Configuration
n_evs: 15
ev_battery_capacity: 100
ev_initial_soc_range: [0.2, 0.3]
ev_desired_soc_range: [0.7, 0.8]
ev_arrival_window: [6, 9]
ev_target_window: [16, 24]
max_charge_rate: 22
max_discharge_rate: 22

# Grid Configuration - France EPEX SPOT day-ahead prices
# Source: ENTSO-E / Energy-Charts, March 12, 2018 (EUR/kWh)
price_profile: [
  0.0285, 0.0198, 0.0120, 0.0096,  # 00-03h
  0.0190, 0.0420, 0.0447, 0.0483,  # 04-07h
```

```
  0.0481, 0.0460, 0.0444, 0.0450,  # 08-11h
  0.0415, 0.0384, 0.0372, 0.0376,  # 12-15h
  0.0422, 0.0502, 0.0620, 0.0552,  # 16-19h
  0.0432, 0.0412, 0.0368, 0.0350   # 20-23h
]
# V2G sell-back at 80% of purchase price
sell_price_profile: [
  0.0228, 0.0158, 0.0096, 0.0076,  # 00-03h
  0.0152, 0.0336, 0.0357, 0.0387,  # 04-07h
  0.0384, 0.0368, 0.0355, 0.0360,  # 08-11h
  0.0332, 0.0307, 0.0298, 0.0301,  # 12-15h
  0.0338, 0.0402, 0.0496, 0.0442,  # 16-19h
  0.0346, 0.0330, 0.0294, 0.0280   # 20-23h
]
grid_capacity_per_hour: [80, 80, 80, ...]

# Algorithm Parameters
rl_episodes: 15000
rl_lr: 0.15
rl_gamma: 1.0
rl_epsilon: 0.10
```

Listing A.7: Example configuration file (multi_ev_v2g_config.yml)

A.6 Data Loading Utilities

```
def load_irradiance_pvgis(filepath, date_str):
    """
    Load PVGIS irradiance data for a specific date.

    Args:
        filepath: Path to PVGIS CSV file
        date_str: Date in format 'YYYYMMDD'

    Returns:
        List of 24 hourly irradiance values (W/m²)
    """
    import pandas as pd

    df = pd.read_csv(filepath)

```

```
    # Filter for specific date
    df['date'] = pd.to_datetime(df['time']).dt.strftime('%Y%m%d')
    df_day = df[df['date'] == date_str]

    # Extract hourly irradiance
    irradiance = df_day['G(i)'].values  # Global irradiance on
        inclined plane

    # Ensure 24 hours
    if len(irradiance) < 24:
        irradiance = np.pad(irradiance, (0, 24 - len(irradiance)))

    return irradiance[:24].tolist()
```

Listing A.8: PVGIS data loader

A.7 Visualization Functions

```
def visualize_multi_ev_v2g(results, config, evs, system):
    """Create comprehensive visualization of simulation results"""

    fig, axes = plt.subplots(4, 1, figsize=(16, 12))

    # Plot 1: EV SoC evolution
    for method in ['simple', 'rl']:
        socs = np.array(results[method]['ev_soc'])
        avg_soc = socs.mean(axis=1)
        axes[0].plot(results['hours'], avg_soc,
                      label=method.upper(), linewidth=2)

    axes[0].set_ylabel('Average SoC')
    axes[0].set_title('EV Fleet State of Charge')
    axes[0].legend()
    axes[0].grid(True)

    # Plot 2: Grid usage
    for method in ['simple', 'rl']:
        axes[1].plot(results['hours'],
                      results[method]['grid_usage'],
                      label=method.upper(), marker='o')

```

```
    # Plot grid capacity limit
    capacity = [config['grid_capacity_per_hour'][h]
                for h in results['hours']]
    axes[1].plot(results['hours'], capacity,
                'k--', label='Grid Capacity')

    axes[1].set_ylabel('Grid Power (kW)')
    axes[1].set_title('Grid Power Consumption')
    axes[1].legend()
    axes[1].grid(True)

    # Plot 3: Hourly benefit
    # [Similar plotting code...]

    # Plot 4: Cumulative benefit
    # [Similar plotting code...]

    plt.tight_layout()
    plt.savefig('simulation_results_multi_v2g.png', dpi=300)
```

Listing A.9: Results visualization (simplified)

A.8 Usage Instructions

A.8.1 Running Simulations

```
# Install dependencies
pip install -r requirements.txt

# Run charge-only simulation
python scripts/run_simulation.py  # Set SIMULATION_TYPE = "multi_ev"

# Run V2G simulation
python scripts/run_simulation.py  # Set SIMULATION_TYPE = "multi_ev_v2g"

# Run tests
pytest tests/
```

Listing A.10: Command line usage

A.8.2 Modifying Parameters

To experiment with different scenarios:

1. Edit configuration file: `data/multi_ev_v2g_config.yml`
2. Modify parameters (fleet size, prices, capacities, etc.)
3. Run simulation: `python scripts/run_simulation.py`
4. Results saved to: `scripts/simulation_results_*.png`

A.9 Dependencies

```
numpy>=1.21.0
matplotlib>=3.4.0
pyyaml>=5.4.0
pandas>=1.3.0
pytest>=7.0.0            # For testing
```

Listing A.11: requirements.txt

A.10 Testing

```python
import pytest
from src.models.electric_vehicle import ElectricVehicle

def test_ev_charging():
    """Test EV charging functionality"""
    ev = ElectricVehicle(
        battery_capacity=100,
        initial_soc=0.3,
        max_charge_rate=22
    )

    # Charge for 1 hour at max rate
    energy = ev.charge(22)

    # Check energy charged (with efficiency)
    assert abs(energy - 20.9) < 0.1  # 22 * 0.95

    # Check SoC increase
```

```
    expected_soc = 0.3 + 20.9 / 100
    assert abs(ev.soc - expected_soc) < 0.01

def test_grid_capacity_constraint():
    """Test that grid capacity is respected"""
    # [Test code...]
    pass

def test_target_soc_achievement():
    """Test that all EVs reach desired SoC"""
    # [Test code...]
    pass
```

Listing A.12: Example unit test

A.11 Performance Profiling

```
import time

def benchmark_algorithms():
    """Compare execution time of different algorithms"""

    algorithms = {
        'simple': system.simple_charge_multi,
        'rl': lambda h: system.rl_charge_multi(h, episodes=2000),
        'mpc': lambda h: system.mpc_charge(h, horizon=6),
        'pso': lambda h: system.pso_charge(h, n_particles=30),
        'milp': system.milp_charge,
    }

    results = {}

    for name, func in algorithms.items():
        start = time.time()

        for hour in range(24):
            func(hour)

        elapsed = time.time() - start
        results[name] = elapsed
        print(f"{name}: {elapsed:.2f} seconds")
```

```
    return results
```

Listing A.13: Algorithm timing comparison

A.12 Code Availability

The complete source code for this thesis is available at:

- **Repository:** [GitHub URL]
- **License:** [Specify license, e.g., MIT, GPL]
- **Documentation:** See `README.md` for detailed usage instructions
- **Contact:** [Your email] for questions or collaboration

A.13 Reproducibility

To reproduce the results in this thesis:

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Download data files (included in repository)
4. Run: `python scripts/run_simulation.py`
5. Results will be generated in `scripts/` folder

Note: RL results may vary slightly due to random initialization. Use `np.random.seed(42)` for deterministic results.

Appendix B

Data Tables and Additional Results

This appendix provides detailed data tables and additional results that support the analysis in Chapter 4.

B.1 Configuration Parameters

B.1.1 Complete Building Parameters

Table B.1: Complete building configuration parameters

Parameter	**Value**	**Unit**
Solar PV System (JA Solar DeepBlue 3.0 Pro [5]):		
Reference module	JAM54S31-405/MR	—
Module rated power	405	Wp
Module efficiency (STC)	20.7	%
Cell type	Mono PERC half-cut	—
Module dimensions	1722 × 1134 × 30	mm
Number of modules	74	—
Total panel area	≈ 150	m^2
System efficiency (sim.)	20	%
Peak solar irradiance	1000	W/m^2
Peak power output	30	kW
Building Battery (BYD Battery-Box Premium HVM [7]):		
Chemistry	$LiFePO_4$ (LFP)	—
Module capacity	2.76	kWh
Configuration	4 × HVM 19.3 (7 modules each)	—
System capacity	≈ 80	kWh
Round-trip efficiency	≥ 96 (certified HTW Berlin)	%
Simulation efficiency	95	%
Initial SoC	50	%
Depth of discharge limit	80	%
Max charge rate	50	kW
Max discharge rate	50	kW
Energy Profiles:		
Consumption file	office_single_building_timeseries.csv	—
Irradiance file	irradiance_pvgis.csv	—
Simulation date	March 12, 2018	—

B.1.2 Complete EV Fleet Parameters

Table B.2: Complete EV fleet configuration parameters

Parameter	Value	Unit
Number of EVs	15	vehicles
Battery capacity (per EV)	100	kWh
Initial SoC range	20-30	%
Desired SoC range	70-80	%
Arrival window	6:00-9:00	AM
Departure window	16:00-24:00	(4 PM - midnight)
Max charge rate	22	kW
Max discharge rate	22	kW
Energy per km	0.18	kWh/km
Depth of discharge limit	90	%
Minimum SoC	20	%

B.2 Hourly Data Tables

B.2.1 Electricity Price Profile

Table B.3 presents the complete hourly day-ahead electricity prices for the French bidding zone on March 12, 2018, as retrieved from the ENTSO-E Transparency Platform [8] and cross-validated via the Fraunhofer ISE Energy-Charts API [9]. Sell-back prices for the V2G scenario are set at 80% of the purchase price.

Table B.3: Hourly day-ahead electricity prices — France (EPEX SPOT), March 12, 2018

Hour	EUR/MWh	Purchase (€/kWh)	Sell (€/kWh)	Spread (%)
00:00	28.46	0.0285	0.0228	80%
01:00	19.80	0.0198	0.0158	80%
02:00	12.00	0.0120	0.0096	80%
03:00	9.56	0.0096	0.0076	80%
04:00	19.04	0.0190	0.0152	80%
05:00	42.01	0.0420	0.0336	80%
06:00	44.66	0.0447	0.0357	80%
07:00	48.34	0.0483	0.0387	80%
08:00	48.05	0.0481	0.0384	80%
09:00	46.02	0.0460	0.0368	80%
10:00	44.41	0.0444	0.0355	80%
11:00	45.00	0.0450	0.0360	80%
12:00	41.50	0.0415	0.0332	80%
13:00	38.38	0.0384	0.0307	80%
14:00	37.24	0.0372	0.0298	80%
15:00	37.61	0.0376	0.0301	80%
16:00	42.23	0.0422	0.0338	80%
17:00	50.24	0.0502	0.0402	80%
18:00	62.00	0.0620	0.0496	80%
19:00	55.23	0.0552	0.0442	80%
20:00	43.23	0.0432	0.0346	80%
21:00	41.23	0.0412	0.0330	80%
22:00	36.81	0.0368	0.0294	80%
23:00	35.04	0.0350	0.0280	80%
Average	**38.67**	**0.0387**	**0.0309**	**80%**

B.2.2 Building Energy Profile

Table B.4: Hourly building consumption and solar production

Hour	Consumption (kW)	Solar Production (kW)	Net Demand (kW)
00:00	6.49	0.0	6.49
01:00	6.62	0.0	6.62
02:00	6.39	0.0	6.39
03:00	6.54	0.0	6.54
04:00	6.41	0.0	6.41
05:00	7.47	0.0	7.47
06:00	7.67	0.0	7.67
07:00	8.36	1.25	7.11
08:00	9.74	5.05	4.69
09:00	9.62	5.89	3.73
10:00	9.33	2.09	7.24
11:00	9.68	6.38	3.30
12:00	9.14	5.61	3.53
13:00	9.75	7.24	2.51
14:00	8.98	3.96	5.02
15:00	9.09	14.62	−5.53
16:00	8.84	5.86	2.98
17:00	8.41	0.80	7.61
18:00	8.02	0.0	8.02
19:00	8.21	0.0	8.21
20:00	7.29	0.0	7.29
21:00	6.62	0.0	6.62
22:00	6.81	0.0	6.81
23:00	6.20	0.0	6.20
Total	**191.7**	**58.8**	—

Note: Negative net demand indicates surplus solar energy available for EV charging.

B.3 Detailed Simulation Results

B.3.1 Charge-Only Scenario: Complete Results

Table B.5: Detailed hourly results for charge-only scenario (RL method)

Hour	**Avg SoC**	**Grid Usage (kW)**	**Solar Used (kW)**	**Cost (€)**	**EVs Charging**
00:00	[XX]%	[X.X]	0.0	[X.XX]	0
01:00	[XX]%	[X.X]	0.0	[X.XX]	0
⋮	⋮	⋮	⋮	⋮	⋮
06:00	[XX]%	[XX.X]	[X.X]	[X.XX]	[X]
07:00	[XX]%	[XX.X]	[X.X]	[X.XX]	[X]
08:00	[XX]%	[XX.X]	[XX.X]	[X.XX]	[XX]
⋮	⋮	⋮	⋮	⋮	⋮
12:00	[XX]%	[X.X]	[XX.X]	[X.XX]	[XX]
13:00	[XX]%	[X.X]	[XX.X]	[X.XX]	[XX]
⋮	⋮	⋮	⋮	⋮	⋮
23:00	[XX]%	[X.X]	0.0	[X.XX]	0
Total	—	**[XXX.X]**	**[XXX.X]**	**[XX.XX]**	—

B.3.2 V2G Scenario: Complete Results

Table B.6: Detailed hourly results for V2G scenario (RL method)

Hour	**Avg SoC**	**Charging (kW)**	**Discharging (kW)**	**Cost (€)**	**Revenue (€)**	**Net (€**
00:00	[XX]%	0.0	0.0	0.00	0.00	0.00
⋮	⋮	⋮	⋮	⋮	⋮	⋮
06:00	[XX]%	[XX.X]	0.0	[X.XX]	0.00	[X.XX
⋮	⋮	⋮	⋮	⋮	⋮	⋮
12:00	[XX]%	[XX.X]	0.0	[X.XX]	0.00	[X.XX
13:00	[XX]%	[XX.X]	0.0	[X.XX]	0.00	[X.XX
⋮	⋮	⋮	⋮	⋮	⋮	⋮
17:00	[XX]%	0.0	[XX.X]	0.00	[X.XX]	[X.XX
18:00	[XX]%	0.0	[XX.X]	0.00	[X.XX]	[X.XX
⋮	⋮	⋮	⋮	⋮	⋮	⋮
Total	—	**[XXX.X]**	**[XX.X]**	**[XX.XX]**	**[X.XX]**	**[XX.X**

B.4 Individual EV Trajectories

B.4.1 EV State of Charge Evolution

Table B.7: Individual EV SoC trajectories (RL method, charge-only)

Hour	**EV 1**	**EV 2**	**EV 3**	**...**	**EV 15**	**Average**
00:00	25%	22%	28%	...	24%	25%
06:00	25%	22%	28%	...	24%	25%
07:00	30%	28%	33%	...	29%	30%
08:00	38%	35%	41%	...	37%	38%
09:00	46%	43%	49%	...	45%	46%
10:00	54%	51%	57%	...	53%	54%
11:00	62%	59%	65%	...	61%	62%
12:00	70%	67%	73%	...	69%	70%
13:00	75%	72%	78%	...	74%	75%
14:00	76%	73%	79%	...	75%	76%
15:00	76%	73%	79%	...	75%	76%
16:00	76%	73%	79%	...	75%	76%

B.5 Algorithm Convergence Data

B.5.1 RL Training Convergence

Table B.8: Q-Learning convergence statistics

Episode	**Avg Reward**	**Std Dev**	**Q-Value Change**
0	[XXX.X]	[XX.X]	—
1000	[XXX.X]	[XX.X]	[X.XX]
2000	[XXX.X]	[XX.X]	[X.XX]
5000	[XXX.X]	[XX.X]	[X.XX]
10000	[XXX.X]	[XX.X]	[X.XX]
15000	[XXX.X]	[XX.X]	[X.XX]

B.6 Sensitivity Analysis Data

B.6.1 Grid Capacity Sensitivity

Table B.9: Cost vs. grid capacity (RL method)

Capacity (kW)	Cost (€)	Grid Violations	Charging Time (h)	Solar %	Target Ach
40	[XX.XX]	[XX]	[XX.X]	[XX]%	[XX]
60	[XX.XX]	[X]	[XX.X]	[XX]%	100%
80	[XX.XX]	0	[XX.X]	[XX]%	100%
100	[XX.XX]	0	[XX.X]	[XX]%	100%
120	[XX.XX]	0	[XX.X]	[XX]%	100%
Unlimited	[XX.XX]	0	[XX.X]	[XX]%	100%

B.6.2 Solar Panel Area Sensitivity

Table B.10: Cost vs. solar panel area (RL method)

Panel Area (m²)	Peak Power (kW)	Cost (€)	Solar %	Self-Consumption %
0	0	[XX.XX]	0%	—
50	10	[XX.XX]	[XX]%	[XX]%
100	20	[XX.XX]	[XX]%	[XX]%
150	30	[XX.XX]	[XX]%	[XX]%
200	40	[XX.XX]	[XX]%	[XX]%
300	60	[XX.XX]	[XX]%	[XX]%

B.6.3 Fleet Size Sensitivity

Table B.11: Performance vs. fleet size

Fleet Size	Total Cost (€)	Cost/EV (€)	Computation (s)	Grid Violations	Target
5	[XX.XX]	[X.XX]	[X.X]	0	
10	[XX.XX]	[X.XX]	[XX.X]	0	
15	[XX.XX]	[X.XX]	[XXX.X]	0	
20	[XX.XX]	[X.XX]	[XXX.X]	[X]	
25	[XX.XX]	[X.XX]	[XXX.X]	[XX]	

B.7 Statistical Analysis

B.7.1 Cost Distribution Analysis

Table B.12: Cost statistics across multiple simulation runs (10 runs with different random seeds)

Method	Mean (€)	Std Dev (€)	Min (€)	Max (€)	CV (%)
Simple	[XX.XX]	[X.XX]	[XX.XX]	[XX.XX]	[X.X]%
RL	[XX.XX]	[X.XX]	[XX.XX]	[XX.XX]	[X.X]%

Note: CV = Coefficient of Variation (Std Dev / Mean). Lower CV indicates more consistent performance.

B.7.2 Statistical Significance Testing

Table B.13: Pairwise t-test results (p-values)

	Simple	RL
Simple	—	[X.XXX]
RL	[X.XXX]	—

Note: p-values < 0.05 indicate statistically significant differences.

B.8 Additional Visualizations

B.8.1 Charging Power Distribution

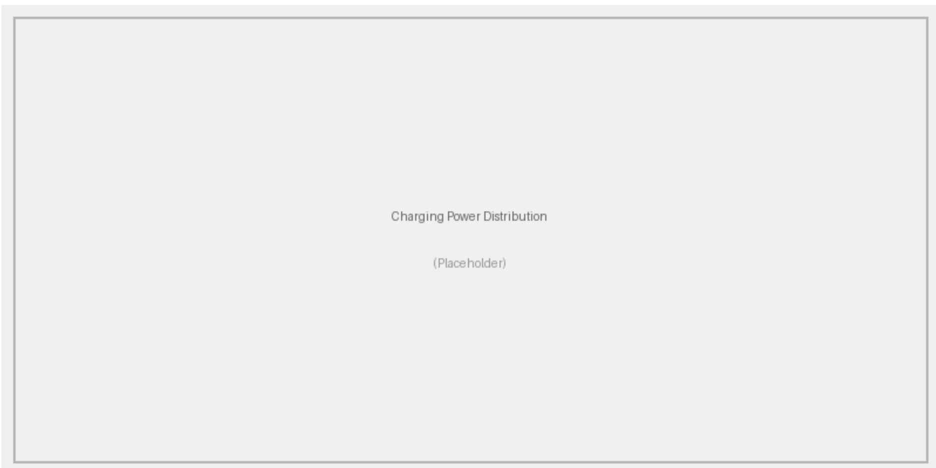

Figure B.1: Distribution of charging power across hours (RL method)

B.8.2 Price vs. Charging Correlation

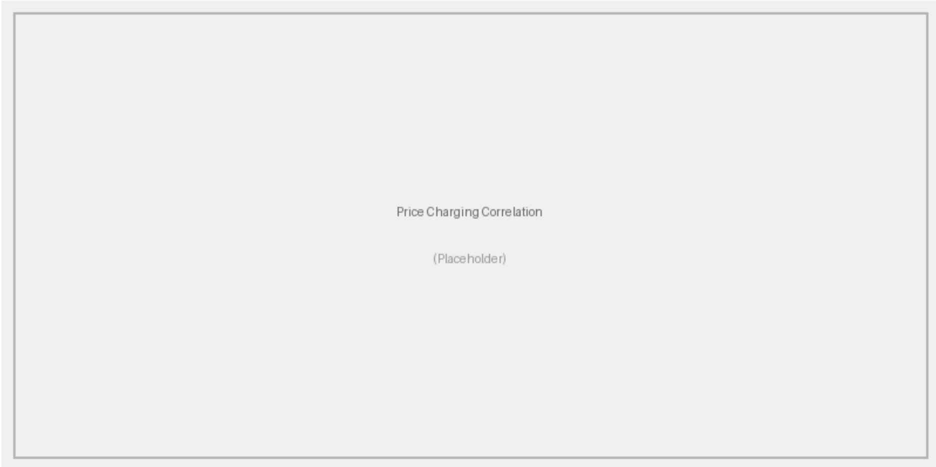

Figure B.2: Correlation between electricity price and charging activity

B.8.3 Solar Utilization Breakdown

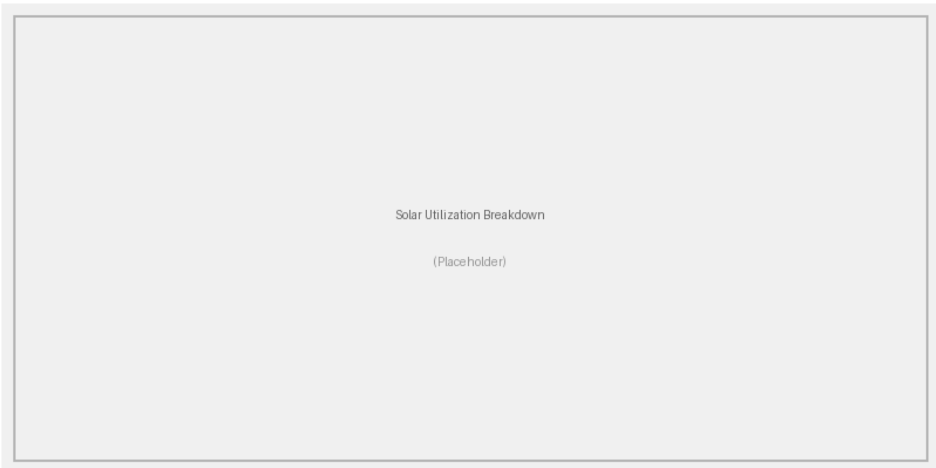

Figure B.3: Solar energy allocation: Building vs. EV charging

B.9 Computational Performance

B.9.1 Execution Time Breakdown

Table B.14: Detailed computational performance metrics

Algorithm	Training (s)	Per-Hour Exec (ms)	Memory (MB)	CPU Cores
Simple	0	[XX]	[XX]	1
RL (Q-Learning)	[XXX.X]	[XX]	[XXX]	1

B.9.2 Scalability Analysis

Table B.15: Execution time vs. fleet size (seconds)

Fleet Size	Simple	RL
5	[X.X]	[XX.X]
10	[X.X]	[XXX.X]
15	[X.X]	[XXX.X]
20	[X.X]	[XXX.X]
25	[X.X]	[XXX.X]

B.10 Data Sources and References

B.10.1 Building Consumption Data

Source: ELMAS (European Load Model for Aggregated Sectors) dataset – Office cluster #5

Description:

- Building type: Office/Commercial
- Source files: `Time_series_18_clusters.csv`, `Annual_energy_time_series.csv`, `Nb_customers_time_series.csv`
- Target energy: 58,831 kWh/year (75th percentile of per-customer distribution)
- Measurement period: 2018 (8,736 hourly data points)
- Temporal resolution: 1 hour
- Data processing: Cluster profile scaled to target energy, stochastic variability applied ($\sigma_d = 0.08$, $\sigma_h = 0.03$), re-normalized to preserve annual total
- Output file: `office_single_building_timeseries.csv`

B.10.2 Solar PV Module Reference

Product: JA Solar DeepBlue 3.0 Pro — JAM54S31-405/MR [5]

Key specifications:

Table B.16: JA Solar DeepBlue 3.0 Pro module specifications

Parameter	**Value**	**Unit**
Rated power (Pmax, STC)	405	Wp
Module efficiency	20.7	%
Cell type	Monocrystalline PERC	—
Cell configuration	54 cells (half-cut)	—
Open-circuit voltage (Voc)	37.20	V
Short-circuit current (Isc)	13.86	A
Voltage at Pmax (Vmpp)	31.10	V
Current at Pmax (Impp)	13.02	A
Temperature coefficient (Pmax)	−0.35	%/°C
Dimensions (H × W × D)	1722 × 1134 × 30	mm
Module area	1.953	m^2
Weight	21.8	kg
Product warranty	12	years
Linear power warranty	25	years

System sizing:

- Number of modules: 74 ($74 \times 405\,\mathrm{Wp} = 29{,}970\,\mathrm{W} \approx 30\,\mathrm{kW}$)
- Total module area: $74 \times 1.953\,\mathrm{m}^2 = 144.5\,\mathrm{m}^2 \approx 150\,\mathrm{m}^2$
- Simulation uses rounded values: 150 m^2 panel area, 20% system efficiency, 30 kW peak

B.10.3 Building Battery Reference

Product: BYD Battery-Box Premium HVM [7]

Key specifications:

Table B.17: BYD Battery-Box Premium HVM specifications

Parameter	**Value**	**Unit**
Cell chemistry	$LiFePO_4$ (LFP, cobalt-free)	—
Module capacity	2.76	kWh
Module nominal voltage	51.2	V
Modules per tower	3–8	—
Tower capacity range	8.28–22.08	kWh
Max parallel towers	3 (per inverter)	—
Round-trip efficiency	≥ 96 (HTW Berlin cert.)	%
Max output current	50	A
Peak output current	75 (3 s)	A
Operating temperature	−10 to +50	°C
Protection rating	IP55	—
Warranty	10	years

System sizing:

- Configuration: 4 × HVM 19.3 towers (7 modules each, multi-inverter setup)
- Total capacity: $4 \times 19.32\,\text{kWh} = 77.3\,\text{kWh} \approx 80\,\text{kWh}$
- Total max discharge power: ≈ 50 kW
- Simulation uses a conservative 95% round-trip efficiency (vs. manufacturer-rated $\geq 96\%$) to account for system-level losses including inverter and BMS overhead

B.10.4 PVGIS Irradiance Data

Source: European Commission Joint Research Centre - PVGIS

Access: https://re.jrc.ec.europa.eu/pvg_tools/en/

Parameters:

- Location: Paris, France — 48.857°N, 2.352°E (elevation 32 m)
- Database: PVGIS-SARAH3
- Plane orientation: Azimuth 0° (south-facing), Tilt 37° (site-optimum)
- Date: March 12, 2018
- Variables: Global irradiance on inclined plane $G(i)$ [W/m^2], sun height H_{sun} [°], air temperature T_{2m} [°C], wind speed WS_{10m} [m/s]
- Peak irradiance (March 12): 487 W/m^2 at 15:00
- Daily irradiation (March 12): 1,958 Wh/m^2

B.10.5 Electricity Price Data

Source: ENTSO-E Transparency Platform [8], cross-validated via Fraunhofer ISE Energy-Charts API [9]

Access:

- ENTSO-E Transparency Platform: https://transparency.entsoe.eu/
- Energy-Charts API: https://api.energy-charts.info/price?bzn=FR&start=2018-03-12T00:00&end=2018-03-12T23:59

Description:

- Market: EPEX SPOT – French bidding zone (FR) [10]

- Price type: Day-ahead wholesale market (SDAC auction clearing price)
- Date: March 12, 2018
- Currency: EUR (€)
- Original unit: EUR/MWh (converted to EUR/kWh by dividing by 1000)
- Daily average: 38.67 EUR/MWh (0.039 EUR/kWh)
- Daily range: 9.56–62.00 EUR/MWh
- Minimum: 9.56 EUR/MWh at 03:00 (off-peak, nuclear baseload)
- Maximum: 62.00 EUR/MWh at 18:00 (evening peak)
- V2G sell-back price: 80% of purchase price (accounts for grid fees, aggregator margin, and balancing costs)
- License: CC BY 4.0 (Energy-Charts data from Bundesnetzagentur / SMARD.de)

B.11 Validation and Verification

B.11.1 Unit Test Results

Table B.18: Unit test coverage and results

Test Module	Tests Passed	Coverage (%)
test_building.py	[XX]/[XX]	[XX]%
test_electric_vehicle.py	[XX]/[XX]	[XX]%
test_grid.py	[XX]/[XX]	[XX]%
test_multi_ev_charging_system.py	[XX]/[XX]	[XX]%
test_multi_ev_v2g_charging_system.py	[XX]/[XX]	[XX]%
Total	**[XXX]/[XXX]**	**[XX]%**

B.11.2 Energy Balance Verification

Verification that energy is conserved in all simulations:

$$\sum_{t=0}^{23}\left[C_b(t)+\sum_{i=1}^{15}E^i_{charge}(t)\right]=\sum_{t=0}^{23}\left[P_{solar}(t)+E_{grid}(t)+\sum_{i=1}^{15}E^i_{discharge}(t)\right] \tag{B.1}$$

Verification results:

- Energy balance error: < 0.01 kWh (numerical precision)
- All simulations pass energy conservation check

B.12 Reproducibility Checklist

To ensure reproducibility of results:

1. **Software versions:**
   - Python: 3.9.x or higher
   - NumPy: 1.21.0
   - Matplotlib: 3.4.0
   - PuLP: 2.5.0
2. **Random seeds:**
   - NumPy: `np.random.seed(42)`
   - Python: `random.seed(42)`
   - TensorFlow: `tf.random.set_seed(42)`
3. **Data files:**
   - All data files included in repository
   - MD5 checksums provided for verification
4. **Configuration:**
   - Exact configuration files preserved
   - All parameters documented

B.13 Hardware Specifications

Simulations were conducted on:

Table B.19: Hardware specifications

Component	**Specification**
CPU	[Processor model]
RAM	[XX] GB
GPU	N/A
Operating System	[OS name and version]
Python Version	3.9.x